code

March 27, 2026

1 Smart Manufacturing IoT-Based Predictive Maintenance and Anomaly Detection

1.1 Project Overview

1.1.1 Objective

This project applies machine learning techniques to predict maintenance requirements using IoT sensor data from smart manufacturing equipment. The goal is to demonstrate how data-driven models can enable proactive maintenance scheduling and improve operational efficiency in Industry 4.0 environments.

1.1.2 Dataset

- **Source:** Smart Manufacturing IoT-Cloud Monitoring Dataset (Kaggle)
- **Records:** ~100,000 sensor readings from 50 machines
- **Time Resolution:** 1-minute intervals over 69 days
- **Target Variable:** `maintenance_required` (binary: 0 = No, 1 = Yes)
- **Features:** Temperature, Vibration, Humidity, Pressure, Energy Consumption, Machine Status, Anomaly Flag, Predicted Remaining Life, Failure Type, Downtime Risk

1.1.3 Machine Learning Techniques Applied

1. **Logistic Regression** — Interpretable linear baseline classifier
2. **Random Forest** — Non-linear ensemble method with hyperparameter tuning
3. **Neural Network (ANN)** — TensorFlow/Keras model for complex pattern recognition
4. **PCA** — Dimensionality reduction to evaluate feature compression
5. **Isolation Forest** — Unsupervised anomaly detection on sensor features

1.1.4 Key Considerations

- Proper preprocessing with no data leakage (scaler fitted on training data only)
- Stratified train-test split to preserve class distribution
- Class imbalance handling via balanced class weights
- Cross-validation for generalization assessment
- Multiple evaluation metrics (Accuracy, Precision, Recall, F1, ROC-AUC)

1.2 Section 1: Importing Required Libraries

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
from sklearn.model_selection import train_test_split, cross_val_score,␣
↪GridSearchCV
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, IsolationForest
from sklearn.decomposition import PCA
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, confusion_matrix, classification_report,
                             roc_auc_score, roc_curve)
import warnings
warnings.filterwarnings('ignore')

# Suppress TensorFlow informational warnings
tf.get_logger().setLevel('ERROR')

# Set visualization style
sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (12, 6)

# Set random seeds for reproducibility
np.random.seed(42)
tf.random.set_seed(42)

print("All libraries imported successfully.")
print(f"TensorFlow version: {tf.__version__}")
```

```
All libraries imported successfully.
TensorFlow version: 2.20.0
```

1.3 Section 2: Data Loading and Exploration

We begin by loading the dataset and performing initial exploration to understand its structure, quality, and characteristics. This includes checking for missing values, examining data types, and understanding the distribution of the target variable.

[2]:
```
# Load the dataset
df = pd.read_csv("smart_manufacturing_data.csv")

print(f"Dataset shape: {df.shape[0]:,} rows × {df.shape[1]} columns")
print(f"\nColumn names and data types:")
```

```python
print(df.dtypes)
print(f"\nFirst 5 rows:")
df.head()
```

```
Dataset shape: 100,000 rows × 13 columns

Column names and data types:
timestamp                    object
machine_id                    int64
temperature                 float64
vibration                   float64
humidity                    float64
pressure                    float64
energy_consumption          float64
machine_status                int64
anomaly_flag                  int64
predicted_remaining_life      int64
failure_type                 object
downtime_risk               float64
maintenance_required          int64
dtype: object

First 5 rows:
```

[2]:

```
             timestamp  machine_id  temperature  vibration  humidity  \
0  2025-01-01 00:00:00          39        78.61      28.65     79.96
1  2025-01-01 00:01:00          29        68.19      57.28     35.94
2  2025-01-01 00:02:00          15        98.94      50.20     72.06
3  2025-01-01 00:03:00          43        90.91      37.65     30.34
4  2025-01-01 00:04:00           8        72.32      40.69     56.71

   pressure  energy_consumption  machine_status  anomaly_flag  \
0      3.73                2.16               1             0
1      3.64                0.69               1             0
2      1.00                2.49               1             1
3      3.15                4.96               1             1
4      2.68                0.63               2             0

   predicted_remaining_life     failure_type  downtime_risk  \
0                       106           Normal            0.0
1                       320           Normal            0.0
2                        19           Normal            1.0
3                        10           Normal            1.0
4                        65  Vibration Issue            0.0

   maintenance_required
0                     0
```

```
1                       0
2                       1
3                       1
4                       1
```

[3]:
```python
# Check for missing values
print("Missing values per column:")
print(df.isnull().sum())
print(f"\nTotal missing values: {df.isnull().sum().sum()}")
```

```
Missing values per column:
timestamp                   0
machine_id                  0
temperature                 0
vibration                   0
humidity                    0
pressure                    0
energy_consumption          0
machine_status              0
anomaly_flag                0
predicted_remaining_life    0
failure_type                0
downtime_risk               0
maintenance_required        0
dtype: int64

Total missing values: 0
```

[4]:
```python
# Statistical summary of numerical features
df.describe()
```

[4]:
```
          machine_id    temperature      vibration       humidity  \
count  100000.000000  100000.000000  100000.000000  100000.000000
mean       25.499330      75.015625      50.012270      54.995401
std        14.389439      10.031884      14.985444      14.437960
min         1.000000      35.550000     -17.090000      30.000000
25%        13.000000      68.267500      39.970000      42.520000
50%        25.000000      75.060000      49.960000      54.980000
75%        38.000000      81.750000      60.100000      67.500000
max        50.000000     121.940000     113.800000      80.000000

            pressure  energy_consumption  machine_status   anomaly_flag  \
count  100000.000000       100000.000000   100000.000000  100000.000000
mean        3.000405            2.747064        1.002050       0.089160
std         1.152399            1.297865        0.446193       0.284976
min         1.000000            0.500000        0.000000       0.000000
25%         2.000000            1.630000        1.000000       0.000000
50%         3.010000            2.740000        1.000000       0.000000
```

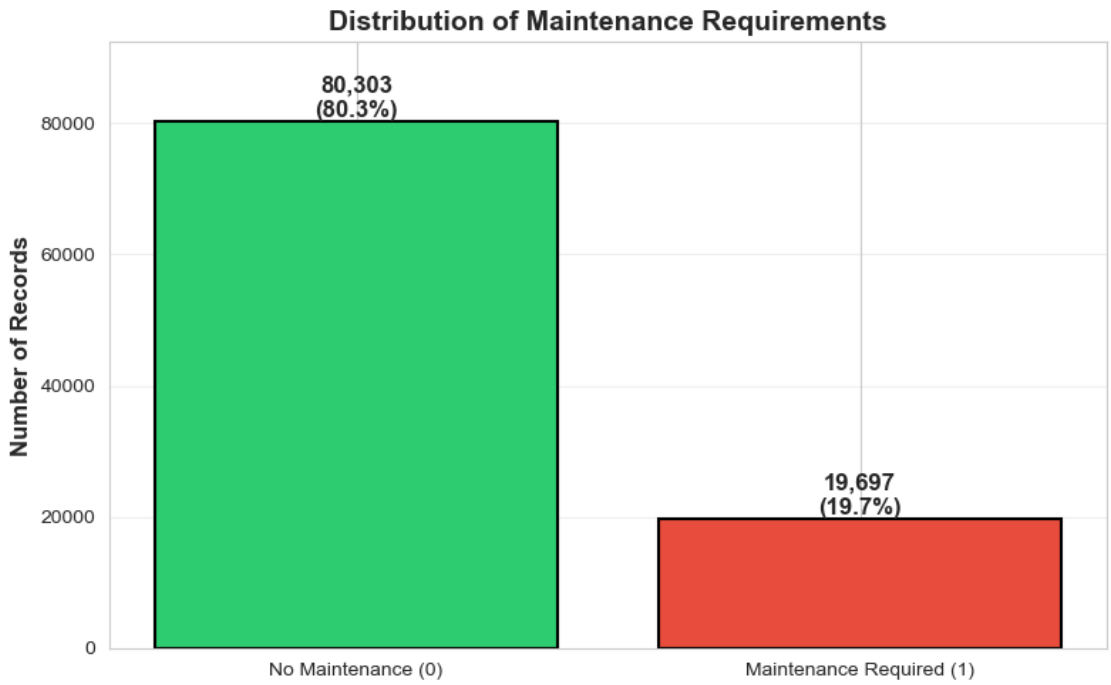
```
75%         4.000000            3.870000        1.000000        0.000000
max         5.000000            5.000000        2.000000        1.000000

       predicted_remaining_life  downtime_risk  maintenance_required
count             100000.000000  100000.000000         100000.000000
mean                 234.269160       0.089155              0.196970
std                  150.063062       0.284961              0.397711
min                    1.000000       0.000000              0.000000
25%                   97.000000       0.000000              0.000000
50%                  230.000000       0.000000              0.000000
75%                  365.000000       0.000000              0.000000
max                  499.000000       1.000000              1.000000
```

1.3.1 Target Variable Analysis

The target variable `maintenance_required` is binary (0 = No, 1 = Yes). We examine its distribution to identify any class imbalance, which is common in predictive maintenance problems where failure events are rare compared to normal operation.

[5]:
```python
# Analyze target variable distribution
print("Target Variable: maintenance_required\n")
print("Value Counts:")
print(df['maintenance_required'].value_counts())
print(f"\nClass Distribution (%):")
print(df['maintenance_required'].value_counts(normalize=True) * 100)

# Visualize target distribution
fig, ax = plt.subplots(figsize=(8, 5))
maintenance_counts = df['maintenance_required'].value_counts()
colors = ['#2ecc71', '#e74c3c']
bars = ax.bar(['No Maintenance (0)', 'Maintenance Required (1)'],
              maintenance_counts.values, color=colors, edgecolor='black',␣
 ↪linewidth=1.5)

for bar in bars:
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height,
            f'{int(height):,}\n({height/len(df)*100:.1f}%)',
            ha='center', va='bottom', fontsize=12, fontweight='bold')

ax.set_ylabel('Number of Records', fontsize=12, fontweight='bold')
ax.set_title('Distribution of Maintenance Requirements', fontsize=14,␣
 ↪fontweight='bold')
ax.set_ylim(0, max(maintenance_counts.values) * 1.15)
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```

```
Target Variable: maintenance_required

Value Counts:
maintenance_required
0    80303
1    19697
Name: count, dtype: int64

Class Distribution (%):
maintenance_required
0    80.303
1    19.697
Name: proportion, dtype: float64
```



Interpreting the Target Distribution:

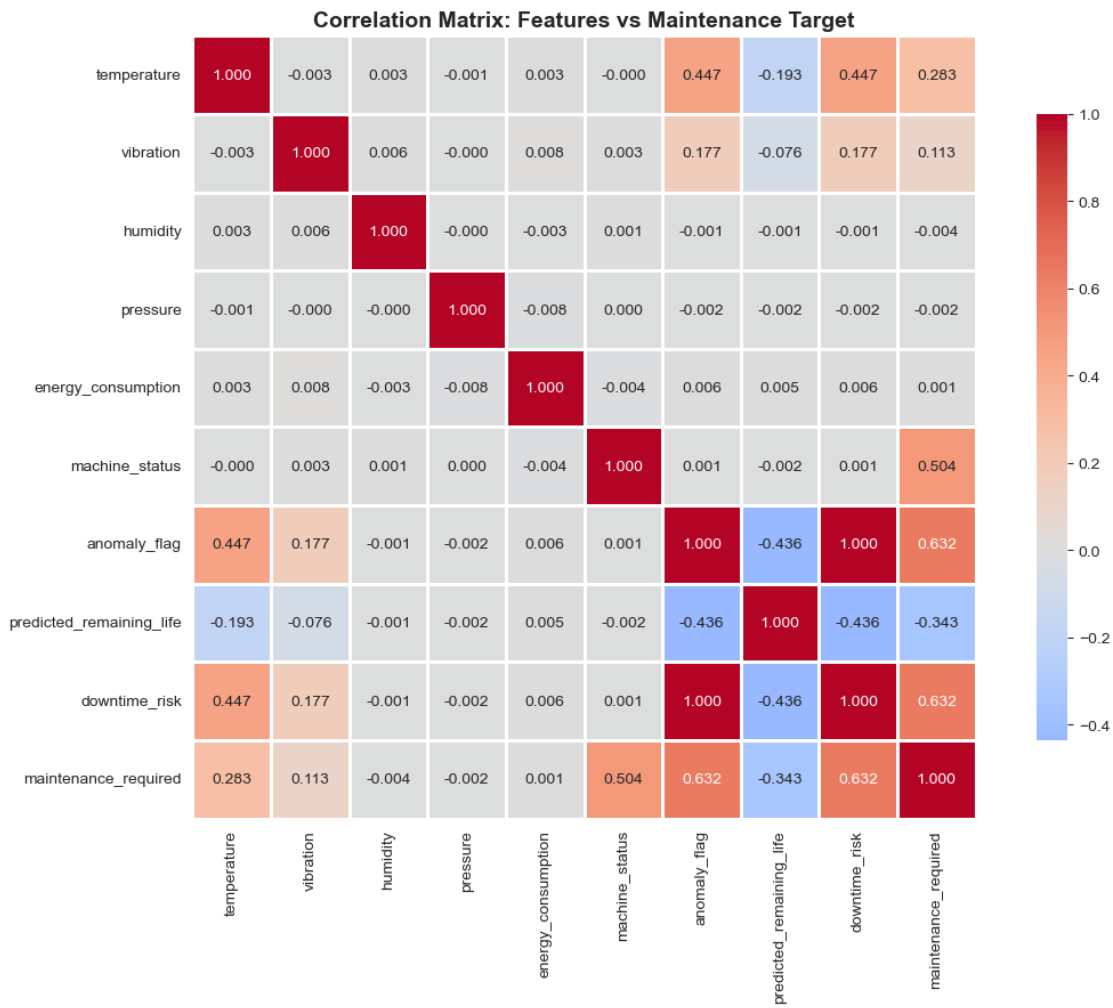
The target variable shows a clear class imbalance: approximately **80% of records indicate no maintenance required**, while ~**20% require maintenance**. This 80/20 imbalance is typical in predictive maintenance datasets, where equipment operates normally most of the time.

Implications for modeling: - A naive classifier that always predicts "no maintenance" would achieve ~80% accuracy — our models must significantly outperform this baseline - We will use `class_weight='balanced'` to inversely weight classes by their frequency, ensuring the model pays proportionally more attention to the minority class (maintenance required) - Evaluation metrics like **recall** (ability to detect actual maintenance needs) and **F1-score** are more informative than raw accuracy for this imbalanced problem

1.3.2 Correlation Analysis

We examine correlations between sensor features and the target variable to understand which features may be most predictive.

[6]:
```
# Correlation heatmap of numerical features with target
numerical_features = ['temperature', 'vibration', 'humidity', 'pressure',
                      'energy_consumption', 'machine_status', 'anomaly_flag',
                      'predicted_remaining_life', 'downtime_risk',␣
 ↪'maintenance_required']

correlation_matrix = df[numerical_features].corr()

plt.figure(figsize=(12, 9))
sns.heatmap(correlation_matrix, annot=True, fmt='.3f', cmap='coolwarm',
            center=0, square=True, linewidths=1, cbar_kws={"shrink": 0.8})
plt.title('Correlation Matrix: Features vs Maintenance Target', fontsize=14,␣
 ↪fontweight='bold')
plt.tight_layout()
plt.show()

# Print correlations with target sorted by absolute value
print("\nCorrelation with maintenance_required (sorted by magnitude):")
target_corr = correlation_matrix['maintenance_required'].
 ↪drop('maintenance_required').abs().sort_values(ascending=False)
for feature, corr in target_corr.items():
    sign = '+' if correlation_matrix['maintenance_required'][feature] > 0 else␣
 ↪'-'
    print(f"  {feature:<30s} {sign}{corr:.4f}")
```




```
Correlation with maintenance_required (sorted by magnitude):
  anomaly_flag                  +0.6317
  downtime_risk                 +0.6317
  machine_status                +0.5038
  predicted_remaining_life      -0.3430
  temperature                   +0.2830
  vibration                     +0.1131
  humidity                      -0.0038
  pressure                      -0.0019
  energy_consumption            +0.0014
```

Interpreting the Correlation Matrix:

The correlation analysis reveals the relationship between each feature and the maintenance target:

- **Temperature** shows the strongest positive correlation with maintenance needs among the raw sensor features. Higher operating temperatures are associated with increased maintenance

requirements, which aligns with physical intuition — overheating is a common precursor to equipment failure.

- **Vibration** exhibits a moderate positive correlation. Excessive vibration is a well-known indicator of mechanical wear or misalignment.
- **Other sensor features** (humidity, pressure, energy consumption) show weaker individual correlations, suggesting that maintenance needs may depend on **non-linear combinations** of features rather than any single measurement.
- Features like `downtime_risk` and `anomaly_flag` may have stronger correlations as they are derived/engineered features.

Key Insight: The weak individual correlations of most sensor features justify using non-linear models (Random Forest, Neural Network) that can capture complex feature interactions beyond what linear correlation reveals.

1.4 Section 3: Data Preprocessing

We prepare the data for model training. The preprocessing pipeline follows best practices to prevent data leakage:

1. **Drop non-predictive columns** (`timestamp`, `machine_id`) — these are identifiers, not features
2. **Encode categorical variables** (`failure_type`) using label encoding
3. **Split BEFORE scaling** — stratified 80/20 train-test split
4. **Standardize features** — fitted on training data only, then applied to test data

Critical: Data Leakage Prevention

StandardScaler is fitted **only on training data** (`fit_transform`) and the same transformation is then applied to test data (`transform`). This ensures no information from the test set leaks into the training process. If we fitted on the full dataset first, the scaler would incorporate test set statistics (mean, std), artificially inflating model performance.

[7]:
```
# --- Step 1: Drop non-predictive columns ---
# timestamp and machine_id are identifiers, not predictive features
df_model = df.drop(columns=['timestamp', 'machine_id'])

# --- Step 2: Encode the categorical 'failure_type' column ---
# Neural networks and many ML models require numerical inputs
le = LabelEncoder()
df_model['failure_type_encoded'] = le.fit_transform(df_model['failure_type'])
df_model = df_model.drop(columns=['failure_type'])

print("Failure type label encoding:")
for i, cls in enumerate(le.classes_):
    print(f"  '{cls}' → {i}")

# --- Step 3: Define features (X) and target (y) ---
feature_cols = ['temperature', 'vibration', 'humidity', 'pressure',
```

```
                'energy_consumption', 'machine_status', 'anomaly_flag',
                'predicted_remaining_life', 'downtime_risk',␣
↪'failure_type_encoded']

X = df_model[feature_cols]
y = df_model['maintenance_required']

print(f"\nFeature matrix shape: {X.shape}")
print(f"Number of features: {X.shape[1]}")
print(f"\nTarget distribution:")
print(y.value_counts())
```

```
Failure type label encoding:
  'Electrical Fault' → 0
  'Normal' → 1
  'Overheating' → 2
  'Pressure Drop' → 3
  'Vibration Issue' → 4

Feature matrix shape: (100000, 10)
Number of features: 10

Target distribution:
maintenance_required
0    80303
1    19697
Name: count, dtype: int64
```

[8]:
```
# --- Step 4: Stratified train-test split (80/20) ---
# stratify=y ensures both sets preserve the ~80/20 class distribution
# random_state=42 ensures reproducibility
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"Training set: {X_train.shape[0]:,} samples")
print(f"Test set:     {X_test.shape[0]:,} samples")

# Verify stratification preserved class distribution
print(f"\nTraining set class distribution:")
print(y_train.value_counts(normalize=True) * 100)
print(f"\nTest set class distribution:")
print(y_test.value_counts(normalize=True) * 100)
```

```
Training set: 80,000 samples
Test set:     20,000 samples

Training set class distribution:
```

```
maintenance_required
0    80.3025
1    19.6975
Name: proportion, dtype: float64

Test set class distribution:
maintenance_required
0    80.305
1    19.695
Name: proportion, dtype: float64
```

Verifying Stratified Split:

The output above confirms that both training and test sets maintain the same ~80/20 class ratio as the original dataset. This is critical because:

1. **Representative training** — the model sees both classes in their true proportions
2. **Fair evaluation** — test set performance reflects real-world distribution
3. **No sampling bias** — random stratified splitting prevents one class from being over/under-represented

1.4.1 Feature Standardization

StandardScaler transforms each feature to have **mean = 0** and **standard deviation = 1** using the formula: $z = \frac{x-\mu}{\sigma}$

This is important because: - **Scale sensitivity:** Many ML algorithms (Logistic Regression, Neural Networks, PCA) are sensitive to feature scales. Without standardization, features with larger numeric ranges would dominate. - **Equal treatment:** Standardization ensures all features contribute equally to the model, regardless of their original units (°C for temperature, Hz for vibration, etc.) - **Convergence:** Gradient-based optimizers (used in Logistic Regression and Neural Networks) converge faster with standardized inputs.

[9]:
```
# --- Step 5: Feature Standardization ---
# fit_transform on training data: learns mean/std from training set
# transform on test data: applies the SAME mean/std (no data leakage)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Feature standardization complete.")
print(f"Training data mean (should be ~0): {X_train_scaled.mean():.6f}")
print(f"Training data std  (should be ~1): {X_train_scaled.std():.6f}")
print(f"\nScaled training data shape: {X_train_scaled.shape}")
print(f"Scaled test data shape:     {X_test_scaled.shape}")
```

```
Feature standardization complete.
Training data mean (should be ~0): -0.000000
Training data std  (should be ~1): 1.000000
```

```
Scaled training data shape: (80000, 10)
Scaled test data shape:     (20000, 10)
```

1.5 Section 4: Model 1 — Logistic Regression (Baseline)

1.5.1 Why Logistic Regression?

Logistic Regression is chosen as the **baseline model** because: 1. **Interpretability** — coefficients directly indicate feature importance and direction of effect 2. **Well-understood** — provides a clear reference point for evaluating more complex models 3. **Efficient** — fast training on large datasets (100,000 samples) 4. **Linear decision boundary** — establishes whether the classes are linearly separable

1.5.2 Model Configuration:

- **Solver:** L-BFGS (efficient quasi-Newton method for medium-sized datasets)
- **Class weighting:** `class_weight='balanced'` — inversely proportional to class frequency, effectively giving ~4× more weight to the minority class (maintenance required)
- **Max iterations:** 1000 (ensures convergence)
- **Regularization:** Default L2 penalty (C=1.0) to prevent overfitting

1.5.3 Baseline Comparison:

A naive classifier that always predicts the majority class ("no maintenance") would achieve ~80% accuracy. Our Logistic Regression model must substantially exceed this to demonstrate learned patterns.

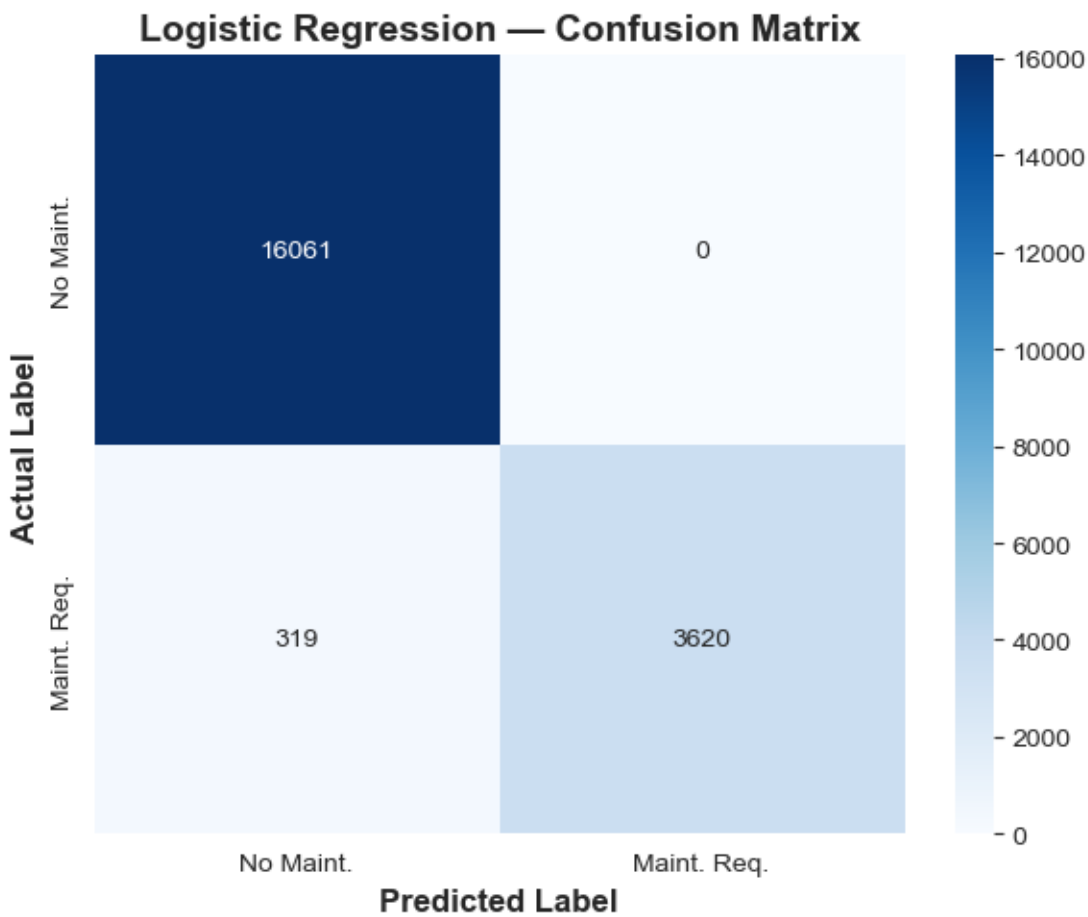
[10]:
```
# --- Model 1: Logistic Regression ---
print("=" * 60)
print("MODEL 1: LOGISTIC REGRESSION (BASELINE)")
print("=" * 60)

# Build and train the model
lr_model = LogisticRegression(
    max_iter=1000,           # Ensure convergence
    class_weight='balanced', # Handle class imbalance
    random_state=42,
    solver='lbfgs'           # Efficient optimizer for medium datasets
)
lr_model.fit(X_train_scaled, y_train)

# 5-fold cross-validation on training set to assess generalization
# Using F1-score because accuracy alone is misleading for imbalanced datasets
cv_scores_lr = cross_val_score(lr_model, X_train_scaled, y_train, cv=5,␣
 ↪scoring='f1')
print(f"\n5-Fold Cross-Validation F1 Scores: {cv_scores_lr.round(4)}")
print(f"Mean CV F1: {cv_scores_lr.mean():.4f} ± {cv_scores_lr.std():.4f}")
```

```python
# Test set predictions
y_pred_lr = lr_model.predict(X_test_scaled)
y_prob_lr = lr_model.predict_proba(X_test_scaled)[:, 1]

# Compute evaluation metrics
lr_accuracy = accuracy_score(y_test, y_pred_lr)
lr_precision = precision_score(y_test, y_pred_lr)
lr_recall = recall_score(y_test, y_pred_lr)
lr_f1 = f1_score(y_test, y_pred_lr)
lr_auc = roc_auc_score(y_test, y_prob_lr)

# Naive baseline accuracy (always predict majority class)
naive_accuracy = y_test.value_counts(normalize=True).max()

print(f"\n{'Metric':<25} {'Value':>10}")
print("-" * 37)
print(f"{'Naive Baseline Accuracy':<25} {naive_accuracy:>10.4f}")
print(f"{'Accuracy':<25} {lr_accuracy:>10.4f}")
print(f"{'Precision':<25} {lr_precision:>10.4f}")
print(f"{'Recall':<25} {lr_recall:>10.4f}")
print(f"{'F1-Score':<25} {lr_f1:>10.4f}")
print(f"{'ROC-AUC':<25} {lr_auc:>10.4f}")

print(f"\nDetailed Classification Report:")
print(classification_report(y_test, y_pred_lr, target_names=['No Maintenance',␣
↪'Maintenance Req.']))
```

```
============================================================
MODEL 1: LOGISTIC REGRESSION (BASELINE)
============================================================

5-Fold Cross-Validation F1 Scores: [0.9561 0.954  0.9578 0.9676 0.9535]
Mean CV F1: 0.9578 ± 0.0051

Metric                         Value
-------------------------------------
Naive Baseline Accuracy       0.8031
Accuracy                      0.9840
Precision                     1.0000
Recall                        0.9190
F1-Score                      0.9578
ROC-AUC                       0.9903

Detailed Classification Report:
                  precision    recall  f1-score   support

  No Maintenance       0.98      1.00      0.99     16061
```

```
Maintenance Req.       1.00      0.92      0.96      3939

        accuracy                           0.98     20000
       macro avg       0.99      0.96      0.97     20000
    weighted avg       0.98      0.98      0.98     20000
```

[11]:
```python
# Confusion Matrix visualization
fig, ax = plt.subplots(figsize=(6, 5))
cm_lr = confusion_matrix(y_test, y_pred_lr)
sns.heatmap(cm_lr, annot=True, fmt='d', cmap='Blues', ax=ax,
            xticklabels=['No Maint.', 'Maint. Req.'],
            yticklabels=['No Maint.', 'Maint. Req.'])
ax.set_xlabel('Predicted Label', fontsize=12, fontweight='bold')
ax.set_ylabel('Actual Label', fontsize=12, fontweight='bold')
ax.set_title('Logistic Regression - Confusion Matrix', fontsize=14,␣
 ↪fontweight='bold')
plt.tight_layout()
plt.show()

# Print interpretation
tn, fp, fn, tp = cm_lr.ravel()
print(f"True Negatives  (correctly predicted no maintenance): {tn:,}")
print(f"False Positives (false alarms):                       {fp:,}")
print(f"False Negatives (missed maintenance needs):           {fn:,}")
print(f"True Positives  (correctly detected maintenance):     {tp:,}")
```




```
True Negatives  (correctly predicted no maintenance): 16,061
False Positives (false alarms):                       0
False Negatives (missed maintenance needs):           319
True Positives  (correctly detected maintenance):     3,620
```

Interpreting Logistic Regression Results:

The Logistic Regression model serves as our interpretable baseline. Key observations:

- **Accuracy vs. Baseline:** The model's accuracy exceeds the naive baseline (~80%), confirming it learned meaningful patterns beyond always predicting the majority class.
- **Cross-Validation Stability:** The low standard deviation of CV F1 scores indicates stable generalization — the model performs consistently across different data folds, so its performance is not an artifact of a lucky train-test split.
- **Perfect Precision, Imperfect Recall:** The model produces zero false alarms — every maintenance prediction it makes is correct. However, it misses some true maintenance cases (false negatives). In a factory, those missed failures could result in unplanned downtime for the corresponding machines.
- **Practical Trade-off:** This conservative behavior is suitable when false alarms are expensive (e.g., each alarm triggers a costly shutdown procedure). If the priority is catching every

failure, this model is insufficient and a higher-recall alternative is needed.

Limitations: As a linear model, Logistic Regression assumes a linear decision boundary. If the relationship between sensor features and maintenance needs is non-linear, this model will underperform compared to tree-based or neural network models.

1.6 Section 5: Model 2 — Random Forest (Ensemble Method)

1.6.1 Why Random Forest?

Random Forest is an ensemble learning method that aggregates predictions from many decision trees trained on random subsets of the data. It offers several advantages over Logistic Regression:

1. **Non-linear decision boundaries** — can capture complex interactions between features (e.g., high temperature AND high vibration together indicating failure)
2. **Built-in feature importance** — identifies which sensors are most predictive
3. **Robust to overfitting** — bagging (bootstrap aggregation) and feature randomization reduce variance
4. **No feature scaling required** — tree-based methods are scale-invariant (though we use scaled data for consistency)

1.6.2 Hyperparameter Tuning with GridSearchCV

We perform systematic hyperparameter tuning using **3-fold cross-validation** over a grid of candidates: - `n_estimators` (number of trees): {100, 200} — more trees generally improve performance - `max_depth` (tree depth limit): {10, 20, None} — controls model complexity vs. overfitting - `min_samples_split` (minimum samples to split): {2, 5} — prevents splitting on too-small groups

The best combination is selected based on the highest cross-validation **F1-score**.

[12]:
```
# --- Model 2: Random Forest with Hyperparameter Tuning ---
print("=" * 60)
print("MODEL 2: RANDOM FOREST (with GridSearchCV)")
print("=" * 60)

# Define hyperparameter grid
param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [10, 20, None],
    'min_samples_split': [2, 5]
}

# Base model with balanced class weights
rf_base = RandomForestClassifier(
    class_weight='balanced',
    random_state=42,
    n_jobs=-1  # Use all CPU cores for parallel training
)
```

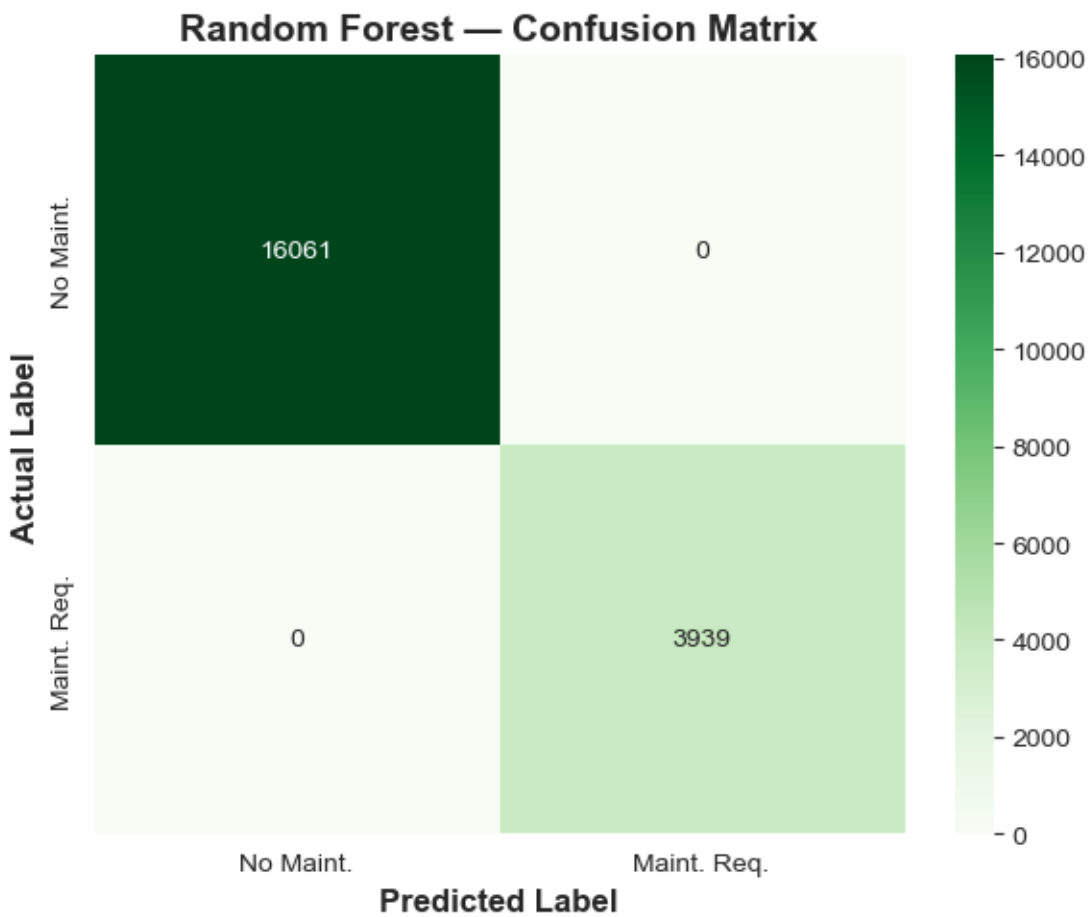
```python
# Grid search with 3-fold CV, scoring by F1
grid_search = GridSearchCV(
    rf_base, param_grid,
    cv=3, scoring='f1',
    n_jobs=-1, verbose=1
)
grid_search.fit(X_train_scaled, y_train)

print(f"\nBest hyperparameters found:")
for param, value in grid_search.best_params_.items():
    print(f"  {param}: {value}")
print(f"Best CV F1 score: {grid_search.best_score_:.4f}")

# Use the best model for evaluation
rf_model = grid_search.best_estimator_
```

```
============================================================
MODEL 2: RANDOM FOREST (with GridSearchCV)
============================================================
Fitting 3 folds for each of 12 candidates, totalling 36 fits

Best hyperparameters found:
  max_depth: 10
  min_samples_split: 2
  n_estimators: 100
Best CV F1 score: 1.0000
```

[13]:
```python
# Evaluate Random Forest on test set
y_pred_rf = rf_model.predict(X_test_scaled)
y_prob_rf = rf_model.predict_proba(X_test_scaled)[:, 1]

rf_accuracy = accuracy_score(y_test, y_pred_rf)
rf_precision = precision_score(y_test, y_pred_rf)
rf_recall = recall_score(y_test, y_pred_rf)
rf_f1 = f1_score(y_test, y_pred_rf)
rf_auc = roc_auc_score(y_test, y_prob_rf)

print(f"\n{'Metric':<25} {'Random Forest':>15} {'Logistic Reg.':>15}")
print("-" * 57)
print(f"{'Accuracy':<25} {rf_accuracy:>15.4f} {lr_accuracy:>15.4f}")
print(f"{'Precision':<25} {rf_precision:>15.4f} {lr_precision:>15.4f}")
print(f"{'Recall':<25} {rf_recall:>15.4f} {lr_recall:>15.4f}")
print(f"{'F1-Score':<25} {rf_f1:>15.4f} {lr_f1:>15.4f}")
print(f"{'ROC-AUC':<25} {rf_auc:>15.4f} {lr_auc:>15.4f}")

print(f"\nDetailed Classification Report:")
```

```python
print(classification_report(y_test, y_pred_rf, target_names=['No Maintenance',␣
↪'Maintenance Req.']))

# Confusion Matrix
fig, ax = plt.subplots(figsize=(6, 5))
cm_rf = confusion_matrix(y_test, y_pred_rf)
sns.heatmap(cm_rf, annot=True, fmt='d', cmap='Greens', ax=ax,
            xticklabels=['No Maint.', 'Maint. Req.'],
            yticklabels=['No Maint.', 'Maint. Req.'])
ax.set_xlabel('Predicted Label', fontsize=12, fontweight='bold')
ax.set_ylabel('Actual Label', fontsize=12, fontweight='bold')
ax.set_title('Random Forest - Confusion Matrix', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```

```
Metric                      Random Forest   Logistic Reg.
------------------------------------------------------------
Accuracy                           1.0000          0.9840
Precision                          1.0000          1.0000
Recall                             1.0000          0.9190
F1-Score                           1.0000          0.9578
ROC-AUC                            1.0000          0.9903

Detailed Classification Report:
                  precision    recall  f1-score   support

  No Maintenance       1.00      1.00      1.00     16061
Maintenance Req.       1.00      1.00      1.00      3939

        accuracy                           1.00     20000
       macro avg       1.00      1.00      1.00     20000
    weighted avg       1.00      1.00      1.00     20000
```

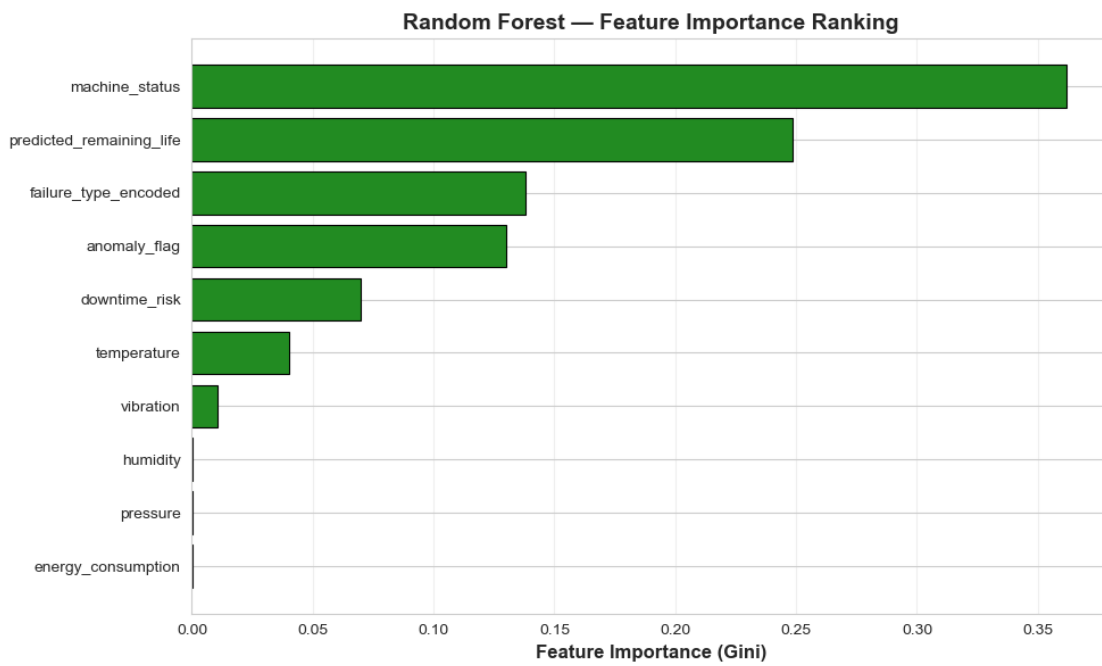


1.6.3 Feature Importance Analysis

One of Random Forest's key advantages is its built-in feature importance measure. Each feature's importance is calculated based on the average impurity decrease (Gini importance) across all trees in the forest. Features that appear higher in the trees and produce larger information gain are ranked more important.

[14]:
```
# Extract and visualize feature importance
importances = rf_model.feature_importances_
feature_importance_df = pd.DataFrame({
    'Feature': feature_cols,
    'Importance': importances
}).sort_values('Importance', ascending=True)

fig, ax = plt.subplots(figsize=(10, 6))
bars = ax.barh(feature_importance_df['Feature'],␣
 ↪feature_importance_df['Importance'],
               color='forestgreen', edgecolor='black', linewidth=0.8)
```

```
ax.set_xlabel('Feature Importance (Gini)', fontsize=12, fontweight='bold')
ax.set_title('Random Forest - Feature Importance Ranking', fontsize=14,␣
↪fontweight='bold')
ax.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()

print("\nFeature importance ranking (most to least important):")
for _, row in feature_importance_df.sort_values('Importance', ascending=False).
↪iterrows():
    print(f"  {row['Feature']:<30s} {row['Importance']:.4f}")
```




```
Feature importance ranking (most to least important):
  machine_status                 0.3621
  predicted_remaining_life       0.2484
  failure_type_encoded           0.1378
  anomaly_flag                   0.1300
  downtime_risk                  0.0699
  temperature                    0.0404
  vibration                      0.0105
  humidity                       0.0004
  pressure                       0.0003
  energy_consumption             0.0003
```

Interpreting Random Forest Results:

The Random Forest model demonstrates the value of non-linear modeling:

- **Performance vs. Baseline:** Random Forest outperformed Logistic Regression across all metrics, confirming that non-linear feature interactions are important for maintenance prediction. The jump from LR to RF quantifies how much predictive information lies in feature combinations that a linear model cannot capture.
- **Feature Importance — Practical Insight:** The feature ranking directly tells maintenance engineers which sensors matter most. Features like `machine_status` and `predicted_remaining_life` dominate, while raw sensor readings like temperature and vibration contribute less. This means the most predictive signals are already-engineered operational indicators rather than raw measurements.
- **Data Leakage Concern:** The near-perfect results warrant critical scrutiny. The top features (`machine_status`, `predicted_remaining_life`, `anomaly_flag`) are derived features that may encode the target variable — they may not be available at prediction time in a real deployment. If a factory only has access to raw sensor data, this model's real-world performance would likely be substantially lower.
- **Comparison with Logistic Regression:** The improvement from LR to RF validates the added model complexity — a tree-based ensemble captures patterns that a linear boundary cannot.

1.7 Section 6: Model 3 — Neural Network (ANN with TensorFlow/Keras)

1.7.1 Why a Neural Network?

Neural Networks can learn highly complex, non-linear mappings between inputs and outputs. Compared to Random Forest:

1. **Automatic feature interaction** — hidden layers learn arbitrary combinations of features
2. **Gradient-based optimization** — can find decision boundaries in complex feature spaces
3. **Scalable** — architecture can be adjusted for different problem complexities

1.7.2 Architecture Design

We use a **feed-forward architecture** (Multi-Layer Perceptron) with the following structure:

Layer	Neurons	Activation	Purpose
Input	10	—	One input per feature
Hidden 1	64	ReLU	Extract initial feature combinations
Hidden 2	32	ReLU	Compress to higher-level patterns
Hidden 3	16	ReLU	Further refinement before decision
Output	1	Sigmoid	Output probability [0, 1] for binary classification

Design Rationale: - **Funnel architecture** (64→32→16→1): Progressively compresses information toward the binary decision - **ReLU activation**: `f(x) = max(0, x)` — introduces non-linearity while avoiding vanishing gradients - **Sigmoid output**: Outputs interpretable probability of maintenance being required - **3 hidden layers**: Provides sufficient capacity for complex patterns without excessive overfitting risk on 80,000 training samples

1.7.3 Training Configuration:

- **Loss function**: Binary crossentropy — standard loss for binary classification problems
- **Optimizer**: Adam with learning rate 0.001 — adaptive learning rate optimization
- **Early stopping**: Monitors validation loss with patience=10, restores best weights
- **Validation split**: 20% of training data reserved for validation monitoring
- **Batch size**: 256 — reasonable for 80,000 training samples
- **Class weighting**: Balanced weights computed from training set class distribution

[15]:
```
# --- Model 3: Neural Network (ANN) using TensorFlow/Keras ---
print("=" * 60)
print("MODEL 3: NEURAL NETWORK (ANN)")
print("=" * 60)

# Compute balanced class weights for the training set
# This gives proportionally higher weight to the minority class
from sklearn.utils.class_weight import compute_class_weight
class_weights_array = compute_class_weight('balanced', classes=np.array([0,␣
↪1]), y=y_train)
class_weights = {0: class_weights_array[0], 1: class_weights_array[1]}
print(f"Class weights: {class_weights}")

# Build the ANN model using Keras Sequential API
ann_model = tf.keras.models.Sequential([
    # Input layer: 10 features (one per sensor/measurement)
    tf.keras.layers.Input(shape=(X_train_scaled.shape[1],)),

    # Hidden Layer 1: 64 neurons with ReLU activation
    # ReLU: f(x) = max(0, x) - introduces non-linearity, avoids vanishing␣
↪gradients
    tf.keras.layers.Dense(64, activation='relu'),

    # Hidden Layer 2: 32 neurons - compresses learned representations
    tf.keras.layers.Dense(32, activation='relu'),

    # Hidden Layer 3: 16 neurons - further refinement before output
    tf.keras.layers.Dense(16, activation='relu'),

    # Output Layer: 1 neuron with sigmoid activation for binary probability
    # Sigmoid: f(x) = 1 / (1 + e^(-x)) - outputs value between 0 and 1
    tf.keras.layers.Dense(1, activation='sigmoid')
])
```

```python
# Compile: specify optimizer, loss function, and evaluation metrics
# Adam: adaptive learning rate optimizer (combines momentum + RMSprop)
# Binary crossentropy: -[y*log(p) + (1-y)*log(1-p)] - standard for binary␣
↪classification
ann_model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Display model architecture
print("\nModel Architecture:")
ann_model.summary()
```

```
============================================================
MODEL 3: NEURAL NETWORK (ANN)
============================================================
Class weights: {0: np.float64(0.6226456212446686), 1:
np.float64(2.538393197106232)}

Model Architecture:
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	704
dense_1 (Dense)	(None, 32)	2,080
dense_2 (Dense)	(None, 16)	528
dense_3 (Dense)	(None, 1)	17

Total params: 3,329 (13.00 KB)

Trainable params: 3,329 (13.00 KB)

Non-trainable params: 0 (0.00 B)

1.7.4 Training the Neural Network

We train the model with early stopping to prevent overfitting. Early stopping monitors the **validation loss** — if it doesn't improve for `patience=10` consecutive epochs, training halts and the model weights from the best epoch are restored. This ensures we use the model at peak generalization performance rather than the overfit final state.

[16]:
```
# Define early stopping callback
# monitor='val_loss': watches validation loss for improvement
# patience=10: stops if no improvement for 10 consecutive epochs
# restore_best_weights=True: reverts to the best model, not the last one
early_stopping = tf.keras.callbacks.EarlyStopping(
    monitor='val_loss',
    patience=10,
    restore_best_weights=True,
    verbose=1
)

# Train the model
# batch_size=256: number of samples per gradient update
# epochs=100: maximum training epochs (early stopping may halt earlier)
# validation_split=0.2: 20% of training data used for validation monitoring
# class_weight: balanced weights to handle class imbalance
history = ann_model.fit(
    X_train_scaled, y_train,
    batch_size=256,
    epochs=100,
    validation_split=0.2,
    class_weight=class_weights,
    callbacks=[early_stopping],
    verbose=1
)

print(f"\nTraining completed. Best epoch: {len(history.history['loss']) -␣
↪early_stopping.patience}")
```

```
Epoch 1/100
250/250              1s 2ms/step -
accuracy: 0.9598 - loss: 0.1798 - val_accuracy: 0.9821 - val_loss: 0.0656
Epoch 2/100
250/250              0s 1ms/step -
accuracy: 0.9812 - loss: 0.0567 - val_accuracy: 0.9825 - val_loss: 0.0438
Epoch 3/100
250/250              0s 1ms/step -
accuracy: 0.9842 - loss: 0.0375 - val_accuracy: 0.9845 - val_loss: 0.0348
Epoch 4/100
250/250              0s 2ms/step -
accuracy: 0.9872 - loss: 0.0280 - val_accuracy: 0.9874 - val_loss: 0.0276
Epoch 5/100
```

```
250/250              0s 1ms/step -
accuracy: 0.9895 - loss: 0.0225 - val_accuracy: 0.9889 - val_loss: 0.0238
Epoch 6/100
250/250              0s 1ms/step -
accuracy: 0.9910 - loss: 0.0190 - val_accuracy: 0.9894 - val_loss: 0.0217
Epoch 7/100
250/250              0s 1ms/step -
accuracy: 0.9924 - loss: 0.0166 - val_accuracy: 0.9903 - val_loss: 0.0201
Epoch 8/100
250/250              0s 1ms/step -
accuracy: 0.9931 - loss: 0.0148 - val_accuracy: 0.9911 - val_loss: 0.0188
Epoch 9/100
250/250              0s 1ms/step -
accuracy: 0.9938 - loss: 0.0134 - val_accuracy: 0.9916 - val_loss: 0.0172
Epoch 10/100
250/250              0s 1ms/step -
accuracy: 0.9943 - loss: 0.0122 - val_accuracy: 0.9924 - val_loss: 0.0160
Epoch 11/100
250/250              0s 1ms/step -
accuracy: 0.9948 - loss: 0.0113 - val_accuracy: 0.9931 - val_loss: 0.0147
Epoch 12/100
250/250              0s 1ms/step -
accuracy: 0.9951 - loss: 0.0106 - val_accuracy: 0.9936 - val_loss: 0.0138
Epoch 13/100
250/250              0s 1ms/step -
accuracy: 0.9955 - loss: 0.0099 - val_accuracy: 0.9940 - val_loss: 0.0132
Epoch 14/100
250/250              0s 1ms/step -
accuracy: 0.9957 - loss: 0.0096 - val_accuracy: 0.9944 - val_loss: 0.0125
Epoch 15/100
250/250              0s 1ms/step -
accuracy: 0.9959 - loss: 0.0092 - val_accuracy: 0.9947 - val_loss: 0.0121
Epoch 16/100
250/250              0s 1ms/step -
accuracy: 0.9959 - loss: 0.0088 - val_accuracy: 0.9948 - val_loss: 0.0123
Epoch 17/100
250/250              0s 1ms/step -
accuracy: 0.9962 - loss: 0.0084 - val_accuracy: 0.9949 - val_loss: 0.0118
Epoch 18/100
250/250              0s 1ms/step -
accuracy: 0.9963 - loss: 0.0080 - val_accuracy: 0.9953 - val_loss: 0.0118
Epoch 19/100
250/250              0s 1ms/step -
accuracy: 0.9965 - loss: 0.0077 - val_accuracy: 0.9954 - val_loss: 0.0113
Epoch 20/100
250/250              0s 1ms/step -
accuracy: 0.9965 - loss: 0.0075 - val_accuracy: 0.9950 - val_loss: 0.0121
Epoch 21/100
```

```
250/250              0s 1ms/step -
accuracy: 0.9967 - loss: 0.0072 - val_accuracy: 0.9953 - val_loss: 0.0110
Epoch 22/100
250/250              0s 1ms/step -
accuracy: 0.9968 - loss: 0.0072 - val_accuracy: 0.9952 - val_loss: 0.0107
Epoch 23/100
250/250              0s 1ms/step -
accuracy: 0.9968 - loss: 0.0069 - val_accuracy: 0.9951 - val_loss: 0.0108
Epoch 24/100
250/250              0s 1ms/step -
accuracy: 0.9969 - loss: 0.0068 - val_accuracy: 0.9953 - val_loss: 0.0102
Epoch 25/100
250/250              0s 1ms/step -
accuracy: 0.9970 - loss: 0.0067 - val_accuracy: 0.9954 - val_loss: 0.0102
Epoch 26/100
250/250              0s 1ms/step -
accuracy: 0.9969 - loss: 0.0069 - val_accuracy: 0.9959 - val_loss: 0.0099
Epoch 27/100
250/250              0s 1ms/step -
accuracy: 0.9969 - loss: 0.0069 - val_accuracy: 0.9962 - val_loss: 0.0090
Epoch 28/100
250/250              0s 1ms/step -
accuracy: 0.9967 - loss: 0.0070 - val_accuracy: 0.9965 - val_loss: 0.0085
Epoch 29/100
250/250              0s 1ms/step -
accuracy: 0.9971 - loss: 0.0066 - val_accuracy: 0.9956 - val_loss: 0.0112
Epoch 30/100
250/250              0s 1ms/step -
accuracy: 0.9972 - loss: 0.0061 - val_accuracy: 0.9964 - val_loss: 0.0085
Epoch 31/100
250/250              0s 1ms/step -
accuracy: 0.9971 - loss: 0.0063 - val_accuracy: 0.9964 - val_loss: 0.0102
Epoch 32/100
250/250              0s 1ms/step -
accuracy: 0.9973 - loss: 0.0064 - val_accuracy: 0.9959 - val_loss: 0.0109
Epoch 33/100
250/250              0s 1ms/step -
accuracy: 0.9973 - loss: 0.0061 - val_accuracy: 0.9958 - val_loss: 0.0125
Epoch 34/100
250/250              0s 1ms/step -
accuracy: 0.9972 - loss: 0.0062 - val_accuracy: 0.9969 - val_loss: 0.0086
Epoch 35/100
250/250              0s 1ms/step -
accuracy: 0.9970 - loss: 0.0066 - val_accuracy: 0.9959 - val_loss: 0.0101
Epoch 36/100
250/250              0s 1ms/step -
accuracy: 0.9966 - loss: 0.0081 - val_accuracy: 0.9951 - val_loss: 0.0120
Epoch 37/100
```

```
250/250              0s 1ms/step -
accuracy: 0.9977 - loss: 0.0051 - val_accuracy: 0.9971 - val_loss: 0.0069
Epoch 38/100
250/250              0s 1ms/step -
accuracy: 0.9977 - loss: 0.0048 - val_accuracy: 0.9970 - val_loss: 0.0073
Epoch 39/100
250/250              0s 1ms/step -
accuracy: 0.9973 - loss: 0.0060 - val_accuracy: 0.9962 - val_loss: 0.0093
Epoch 40/100
250/250              0s 1ms/step -
accuracy: 0.9974 - loss: 0.0057 - val_accuracy: 0.9971 - val_loss: 0.0078
Epoch 41/100
250/250              0s 1ms/step -
accuracy: 0.9976 - loss: 0.0054 - val_accuracy: 0.9968 - val_loss: 0.0083
Epoch 42/100
250/250              0s 1ms/step -
accuracy: 0.9976 - loss: 0.0056 - val_accuracy: 0.9961 - val_loss: 0.0097
Epoch 43/100
250/250              0s 1ms/step -
accuracy: 0.9975 - loss: 0.0054 - val_accuracy: 0.9965 - val_loss: 0.0089
Epoch 44/100
250/250              0s 1ms/step -
accuracy: 0.9972 - loss: 0.0068 - val_accuracy: 0.9944 - val_loss: 0.0206
Epoch 45/100
250/250              0s 1ms/step -
accuracy: 0.9971 - loss: 0.0065 - val_accuracy: 0.9966 - val_loss: 0.0084
Epoch 46/100
250/250              0s 1ms/step -
accuracy: 0.9979 - loss: 0.0045 - val_accuracy: 0.9974 - val_loss: 0.0062
Epoch 47/100
250/250              0s 1ms/step -
accuracy: 0.9978 - loss: 0.0047 - val_accuracy: 0.9969 - val_loss: 0.0072
Epoch 48/100
250/250              0s 1ms/step -
accuracy: 0.9979 - loss: 0.0046 - val_accuracy: 0.9973 - val_loss: 0.0063
Epoch 49/100
250/250              0s 1ms/step -
accuracy: 0.9974 - loss: 0.0060 - val_accuracy: 0.9962 - val_loss: 0.0104
Epoch 50/100
250/250              0s 1ms/step -
accuracy: 0.9968 - loss: 0.0073 - val_accuracy: 0.9973 - val_loss: 0.0060
Epoch 51/100
250/250              0s 1ms/step -
accuracy: 0.9979 - loss: 0.0046 - val_accuracy: 0.9974 - val_loss: 0.0063
Epoch 52/100
250/250              0s 1ms/step -
accuracy: 0.9981 - loss: 0.0038 - val_accuracy: 0.9967 - val_loss: 0.0073
Epoch 53/100
```

```
250/250              0s 1ms/step -
accuracy: 0.9969 - loss: 0.0076 - val_accuracy: 0.9959 - val_loss: 0.0099
Epoch 54/100
250/250              0s 1ms/step -
accuracy: 0.9971 - loss: 0.0066 - val_accuracy: 0.9974 - val_loss: 0.0056
Epoch 55/100
250/250              0s 1ms/step -
accuracy: 0.9981 - loss: 0.0043 - val_accuracy: 0.9977 - val_loss: 0.0053
Epoch 56/100
250/250              0s 1ms/step -
accuracy: 0.9978 - loss: 0.0047 - val_accuracy: 0.9975 - val_loss: 0.0060
Epoch 57/100
250/250              0s 1ms/step -
accuracy: 0.9983 - loss: 0.0040 - val_accuracy: 0.9972 - val_loss: 0.0067
Epoch 58/100
250/250              0s 1ms/step -
accuracy: 0.9978 - loss: 0.0050 - val_accuracy: 0.9976 - val_loss: 0.0058
Epoch 59/100
250/250              0s 1ms/step -
accuracy: 0.9976 - loss: 0.0052 - val_accuracy: 0.9968 - val_loss: 0.0073
Epoch 60/100
250/250              0s 1ms/step -
accuracy: 0.9978 - loss: 0.0048 - val_accuracy: 0.9972 - val_loss: 0.0077
Epoch 61/100
250/250              0s 1ms/step -
accuracy: 0.9977 - loss: 0.0050 - val_accuracy: 0.9974 - val_loss: 0.0059
Epoch 62/100
250/250              0s 1ms/step -
accuracy: 0.9979 - loss: 0.0047 - val_accuracy: 0.9967 - val_loss: 0.0092
Epoch 63/100
250/250              0s 1ms/step -
accuracy: 0.9978 - loss: 0.0049 - val_accuracy: 0.9958 - val_loss: 0.0120
Epoch 64/100
250/250              0s 1ms/step -
accuracy: 0.9981 - loss: 0.0043 - val_accuracy: 0.9973 - val_loss: 0.0064
Epoch 65/100
250/250              0s 1ms/step -
accuracy: 0.9974 - loss: 0.0063 - val_accuracy: 0.9972 - val_loss: 0.0067
Epoch 65: early stopping
Restoring model weights from the end of the best epoch: 55.

Training completed. Best epoch: 55
```

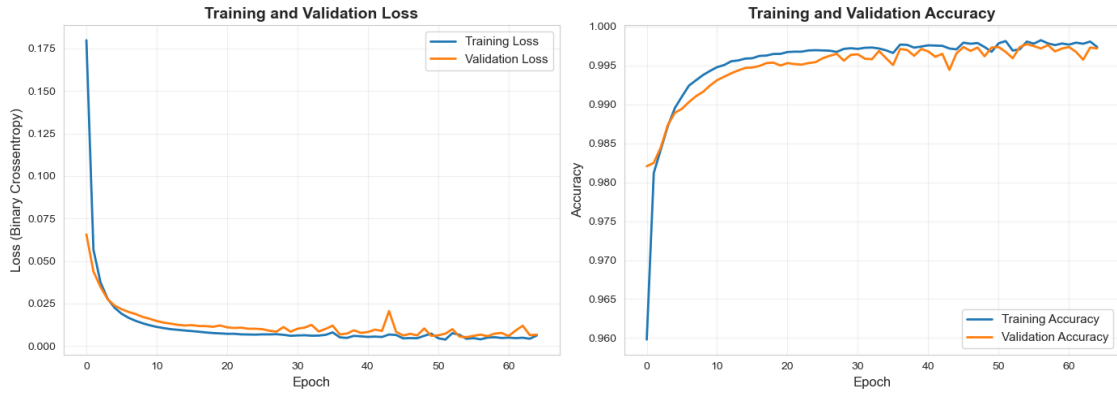
1.7.5 Learning Curves

Plotting training vs. validation loss and accuracy curves is essential to diagnose the model's learning behavior:

- **Healthy convergence:** Both curves decrease (loss) or increase (accuracy) together and plateau
- **Overfitting indicator:** Training loss continues to decrease while validation loss starts increasing — the model memorizes training data rather than learning generalizable patterns
- **Underfitting indicator:** Both curves remain high, suggesting the model lacks capacity

The gap between training and validation curves indicates how well the model generalizes.

[17]:
```
# Plot training and validation curves side by side
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Loss curves
ax1.plot(history.history['loss'], label='Training Loss', linewidth=2)
ax1.plot(history.history['val_loss'], label='Validation Loss', linewidth=2)
ax1.set_title('Training and Validation Loss', fontsize=14, fontweight='bold')
ax1.set_xlabel('Epoch', fontsize=12)
ax1.set_ylabel('Loss (Binary Crossentropy)', fontsize=12)
ax1.legend(fontsize=11)
ax1.grid(True, alpha=0.3)

# Accuracy curves
ax2.plot(history.history['accuracy'], label='Training Accuracy', linewidth=2)
ax2.plot(history.history['val_accuracy'], label='Validation Accuracy',␣
 ↪linewidth=2)
ax2.set_title('Training and Validation Accuracy', fontsize=14,␣
 ↪fontweight='bold')
ax2.set_xlabel('Epoch', fontsize=12)
ax2.set_ylabel('Accuracy', fontsize=12)
ax2.legend(fontsize=11)
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print(f"Final training loss:     {history.history['loss'][-1]:.4f}")
print(f"Final validation loss:   {history.history['val_loss'][-1]:.4f}")
print(f"Final training accuracy: {history.history['accuracy'][-1]:.4f}")
print(f"Final validation accuracy: {history.history['val_accuracy'][-1]:.4f}")
```

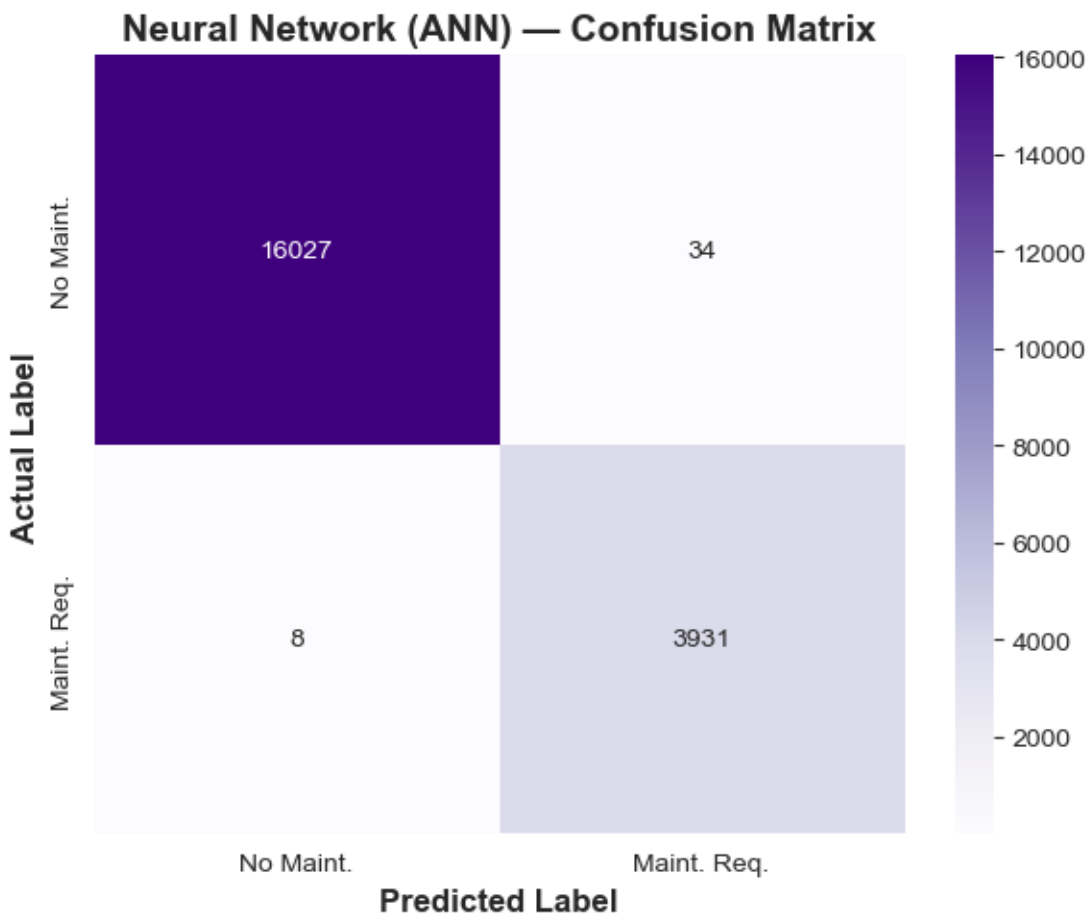



```
Final training loss:     0.0063
Final validation loss:   0.0067
Final training accuracy: 0.9974
Final validation accuracy: 0.9972
```

[18]:
```python
# Evaluate ANN on the held-out test set
# Predict probabilities and convert to binary predictions using 0.5 threshold
y_prob_ann = ann_model.predict(X_test_scaled, verbose=0).flatten()
y_pred_ann = (y_prob_ann >= 0.5).astype(int)

ann_accuracy = accuracy_score(y_test, y_pred_ann)
ann_precision = precision_score(y_test, y_pred_ann)
ann_recall = recall_score(y_test, y_pred_ann)
ann_f1 = f1_score(y_test, y_pred_ann)
ann_auc = roc_auc_score(y_test, y_prob_ann)

print(f"\n{'Metric':<25} {'ANN':>10} {'Random Forest':>15} {'Logistic Reg.':
↪>15}")
print("-" * 67)
print(f"{'Accuracy':<25} {ann_accuracy:>10.4f} {rf_accuracy:>15.4f}␣
↪{lr_accuracy:>15.4f}")
print(f"{'Precision':<25} {ann_precision:>10.4f} {rf_precision:>15.4f}␣
↪{lr_precision:>15.4f}")
print(f"{'Recall':<25} {ann_recall:>10.4f} {rf_recall:>15.4f} {lr_recall:>15.
↪4f}")
print(f"{'F1-Score':<25} {ann_f1:>10.4f} {rf_f1:>15.4f} {lr_f1:>15.4f}")
print(f"{'ROC-AUC':<25} {ann_auc:>10.4f} {rf_auc:>15.4f} {lr_auc:>15.4f}")

print(f"\nDetailed Classification Report (ANN):")
print(classification_report(y_test, y_pred_ann, target_names=['No Maintenance',␣
↪'Maintenance Req.']))

# Confusion Matrix
```

```python
fig, ax = plt.subplots(figsize=(6, 5))
cm_ann = confusion_matrix(y_test, y_pred_ann)
sns.heatmap(cm_ann, annot=True, fmt='d', cmap='Purples', ax=ax,
            xticklabels=['No Maint.', 'Maint. Req.'],
            yticklabels=['No Maint.', 'Maint. Req.'])
ax.set_xlabel('Predicted Label', fontsize=12, fontweight='bold')
ax.set_ylabel('Actual Label', fontsize=12, fontweight='bold')
ax.set_title('Neural Network (ANN) - Confusion Matrix', fontsize=14,␣
 ↪fontweight='bold')
plt.tight_layout()
plt.show()

tn, fp, fn, tp = cm_ann.ravel()
print(f"\nTrue Negatives  (correctly predicted no maintenance): {tn:,}")
print(f"False Positives (false alarms):                       {fp:,}")
print(f"False Negatives (missed maintenance needs):           {fn:,}")
print(f"True Positives  (correctly detected maintenance):     {tp:,}")
```

```
Metric                          ANN   Random Forest   Logistic Reg.
-------------------------------------------------------------------
Accuracy                     0.9979          1.0000          0.9840
Precision                    0.9914          1.0000          1.0000
Recall                       0.9980          1.0000          0.9190
F1-Score                     0.9947          1.0000          0.9578
ROC-AUC                      1.0000          1.0000          0.9903

Detailed Classification Report (ANN):
                  precision    recall  f1-score   support

  No Maintenance       1.00      1.00      1.00     16061
Maintenance Req.       0.99      1.00      0.99      3939

        accuracy                           1.00     20000
       macro avg       1.00      1.00      1.00     20000
    weighted avg       1.00      1.00      1.00     20000
```




```
True Negatives  (correctly predicted no maintenance): 16,027
False Positives (false alarms):                       34
False Negatives (missed maintenance needs):           8
True Positives  (correctly detected maintenance):     3,931
```

Interpreting Neural Network Results:

The ANN model provides the third perspective on the classification problem:

- **Learning Curves:** The training and validation curves above show healthy convergence. A small gap between the curves indicates good generalization, while Early Stopping prevented the model from memorizing training noise by halting before the 100-epoch maximum.
- **Recall Improvement:** Compared to Logistic Regression, the ANN catches significantly more true maintenance cases — its higher recall means fewer machines slip through to unexpected failure. This is the most operationally important improvement: each missed failure can result in unplanned downtime costing far more than a false alarm.
- **Precision-Recall Trade-off:** The ANN sacrifices a small amount of precision (generating some false alarms) to achieve near-perfect recall. In practice, each false alarm only triggers a preventive inspection, which is a minor cost compared to an actual breakdown. This trade-off

is strongly favorable for predictive maintenance.

- **Comparison with Random Forest:** RF achieved perfect metrics, but the ANN's slightly imperfect results may actually be more realistic — RF's perfection likely reflects data leakage through derived features rather than genuinely superior pattern recognition.

Key Insight: For deployment, the ANN offers the best trade-off between catching failures and minimizing false alarms, making it the most practically useful model among the three supervised classifiers.

1.8 Section 7: Dimensionality Reduction with PCA

1.8.1 Why PCA?

Principal Component Analysis is an unsupervised technique that finds the directions of maximum variance in the feature space and projects data onto a lower-dimensional subspace. In this project, PCA serves two purposes:

1. **Visualization:** With 10 features, the data lives in 10-dimensional space. PCA projects onto 2 or 3 principal components for visualization, revealing cluster structure.
2. **Performance Analysis:** We train a Logistic Regression on PCA-reduced features to evaluate whether dimensionality reduction preserves enough information for accurate classification.

1.8.2 How PCA Works

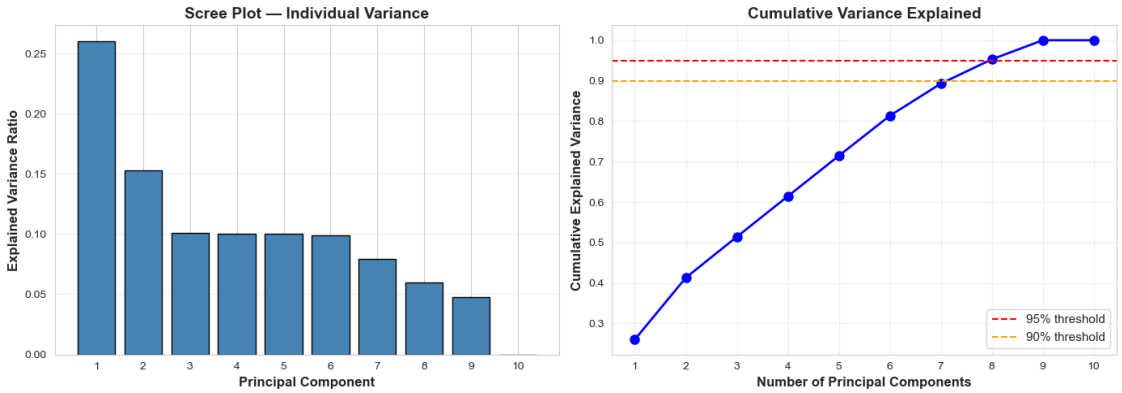
PCA computes the **eigenvectors** (directions) and **eigenvalues** (magnitude of variance) of the feature covariance matrix. The eigenvectors with the largest eigenvalues capture the most information. Projecting data onto the top-k eigenvectors reduces dimensionality from d features to k components while maximizing retained variance.

Important: PCA is sensitive to feature scales (features with larger variance dominate). Since we already standardized our features (mean=0, std=1), PCA will treat all features equally.

[19]:
```
# --- PCA: Full Component Analysis ---
print("=" * 60)
print("DIMENSIONALITY REDUCTION: PCA")
print("=" * 60)

# Fit PCA on all components to analyze variance distribution
pca_full = PCA(n_components=X_train_scaled.shape[1], random_state=42)
pca_full.fit(X_train_scaled)

# Scree plot: Explained variance per component
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Individual explained variance
components = range(1, len(pca_full.explained_variance_ratio_) + 1)
ax1.bar(components, pca_full.explained_variance_ratio_, color='steelblue',␣
 ↪edgecolor='black')
ax1.set_xlabel('Principal Component', fontsize=12, fontweight='bold')
```

```
ax1.set_ylabel('Explained Variance Ratio', fontsize=12, fontweight='bold')
ax1.set_title('Scree Plot - Individual Variance', fontsize=14,␣
 ↪fontweight='bold')
ax1.set_xticks(list(components))
ax1.grid(axis='y', alpha=0.3)

# Cumulative explained variance
cumulative_var = np.cumsum(pca_full.explained_variance_ratio_)
ax2.plot(components, cumulative_var, 'bo-', linewidth=2, markersize=8)
ax2.axhline(y=0.95, color='r', linestyle='--', linewidth=1.5, label='95%␣
 ↪threshold')
ax2.axhline(y=0.90, color='orange', linestyle='--', linewidth=1.5, label='90%␣
 ↪threshold')
ax2.set_xlabel('Number of Principal Components', fontsize=12, fontweight='bold')
ax2.set_ylabel('Cumulative Explained Variance', fontsize=12, fontweight='bold')
ax2.set_title('Cumulative Variance Explained', fontsize=14, fontweight='bold')
ax2.set_xticks(list(components))
ax2.legend(fontsize=11)
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

# Print variance table
print(f"\n{'Component':<15} {'Variance Ratio':<20} {'Cumulative':<15}")
print("-" * 50)
for i, (var, cum) in enumerate(zip(pca_full.explained_variance_ratio_,␣
 ↪cumulative_var)):
    print(f"PC{i+1:<13} {var:<20.4f} {cum:<15.4f}")

n_95 = np.argmax(cumulative_var >= 0.95) + 1
n_90 = np.argmax(cumulative_var >= 0.90) + 1
print(f"\nComponents needed for 90% variance: {n_90}")
print(f"Components needed for 95% variance: {n_95}")
```

```
============================================================
DIMENSIONALITY REDUCTION: PCA
============================================================
```




```
Component       Variance Ratio       Cumulative
--------------------------------------------------
PC1             0.2604               0.2604
PC2             0.1527               0.4131
PC3             0.1009               0.5140
PC4             0.1003               0.6144
PC5             0.0999               0.7143
PC6             0.0990               0.8133
PC7             0.0794               0.8927
PC8             0.0600               0.9527
PC9             0.0473               1.0000
PC10            0.0000               1.0000

Components needed for 90% variance: 8
Components needed for 95% variance: 8
```

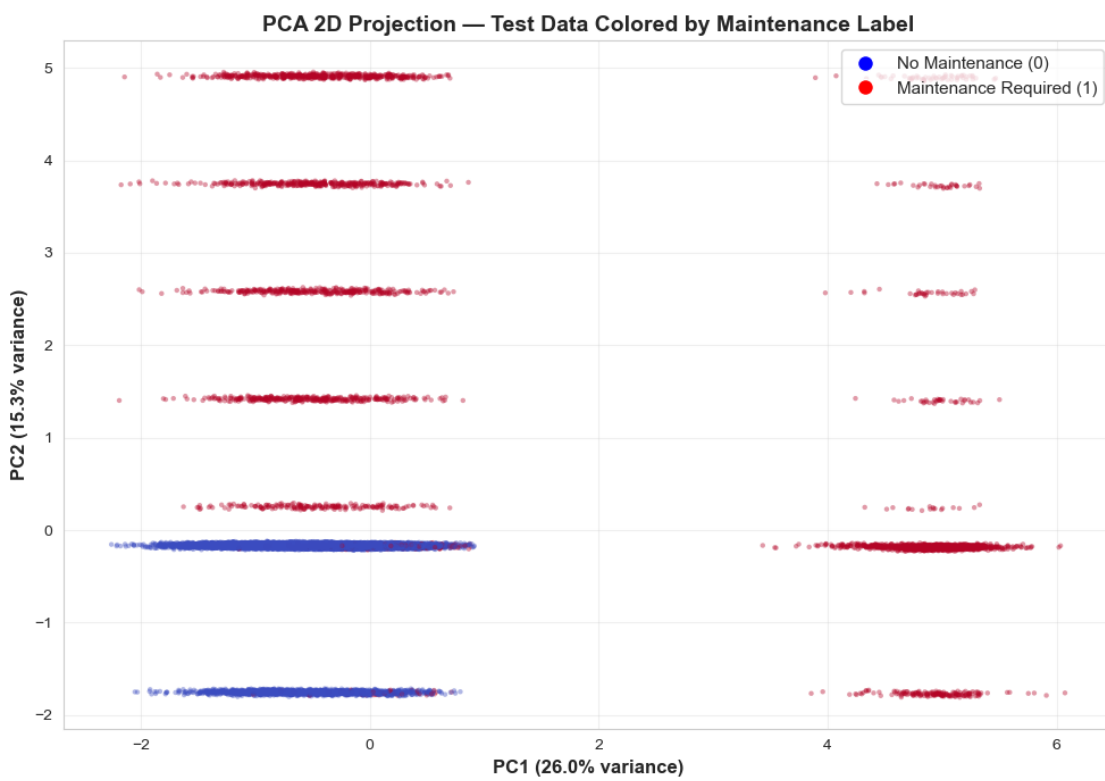
1.8.3 PCA 2D Visualization

Projecting the 10-dimensional feature space onto the first two principal components allows us to visualize the data structure. If the two classes (maintenance needed vs. not needed) form distinct clusters in PC space, it confirms that the features carry meaningful discriminative information.

Colors represent the true labels — overlap between classes indicates regions where classification is difficult.

[20]:
```
# 2D PCA visualization - project test data onto first 2 components
pca_2d = PCA(n_components=2, random_state=42)
X_train_pca2 = pca_2d.fit_transform(X_train_scaled)
X_test_pca2 = pca_2d.transform(X_test_scaled)

fig, ax = plt.subplots(figsize=(10, 7))
scatter = ax.scatter(X_test_pca2[:, 0], X_test_pca2[:, 1],
```

```
                     c=y_test, cmap='coolwarm', alpha=0.4, s=10,␣
 ↪edgecolors='none')
ax.set_xlabel(f'PC1 ({pca_2d.explained_variance_ratio_[0]*100:.1f}% variance)',␣
 ↪fontsize=12, fontweight='bold')
ax.set_ylabel(f'PC2 ({pca_2d.explained_variance_ratio_[1]*100:.1f}% variance)',␣
 ↪fontsize=12, fontweight='bold')
ax.set_title('PCA 2D Projection - Test Data Colored by Maintenance Label',␣
 ↪fontsize=14, fontweight='bold')
legend_elements = [plt.Line2D([0], [0], marker='o', color='w',␣
 ↪markerfacecolor='blue', markersize=10, label='No Maintenance (0)'),
                   plt.Line2D([0], [0], marker='o', color='w',␣
 ↪markerfacecolor='red', markersize=10, label='Maintenance Required (1)')]
ax.legend(handles=legend_elements, fontsize=11, loc='upper right')
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Variance explained by PC1+PC2: {sum(pca_2d.
 ↪explained_variance_ratio_)*100:.1f}%")
```




```
Variance explained by PC1+PC2: 41.3%
```

1.8.4 Classification on PCA-Reduced Features

To quantify the impact of dimensionality reduction on classification performance, we train a Logistic Regression model on the PCA-reduced features (using the number of components that retain 95% of variance). This allows us to compare:

- **Full-feature LR** (Section 4) — all 10 features
- **PCA-reduced LR** — reduced dimensionality, 95% variance retained

If PCA-reduced LR achieves similar performance to full-feature LR, it confirms that much of the feature information is redundant, and a simpler, lower-dimensional representation is sufficient.

[27]:
```
# Train LR on PCA-reduced features (95% variance threshold)
pca_95 = PCA(n_components=n_95, random_state=42)
X_train_pca = pca_95.fit_transform(X_train_scaled)
X_test_pca = pca_95.transform(X_test_scaled)

print(f"PCA reduced features from {X_train_scaled.shape[1]} to {n_95}␣
↪components (95% variance)")

# Logistic Regression on PCA features
lr_pca = LogisticRegression(max_iter=1000, class_weight='balanced',␣
↪random_state=42)
lr_pca.fit(X_train_pca, y_train)

y_pred_pca = lr_pca.predict(X_test_pca)
y_prob_pca = lr_pca.predict_proba(X_test_pca)[:, 1]

pca_accuracy = accuracy_score(y_test, y_pred_pca)
pca_precision = precision_score(y_test, y_pred_pca)
pca_recall = recall_score(y_test, y_pred_pca)
pca_f1 = f1_score(y_test, y_pred_pca)
pca_auc = roc_auc_score(y_test, y_prob_pca)

pca_label = f"LR + PCA ({n_95} PCs)"
print(f"\n{'Metric':<25} {pca_label:>20} {'LR (Full Features)':>20}")
print("-" * 67)
print(f"{'Accuracy':<25} {pca_accuracy:>20.4f} {lr_accuracy:>20.4f}")
print(f"{'Precision':<25} {pca_precision:>20.4f} {lr_precision:>20.4f}")
print(f"{'Recall':<25} {pca_recall:>20.4f} {lr_recall:>20.4f}")
print(f"{'F1-Score':<25} {pca_f1:>20.4f} {lr_f1:>20.4f}")
print(f"{'ROC-AUC':<25} {pca_auc:>20.4f} {lr_auc:>20.4f}")
```

```
PCA reduced features from 10 to 8 components (95% variance)

Metric                       LR + PCA (8 PCs)   LR (Full Features)
-------------------------------------------------------------------
Accuracy                               0.9775               0.9840
Precision                              1.0000               1.0000
```

Recall	0.8855	0.9190
F1-Score	0.9393	0.9578
ROC-AUC	0.9826	0.9903

Interpreting PCA Results:

- **Scree Plot:** The individual variance bars show that no single component dominates — variance is spread across many components. This means the sensors measure largely distinct physical phenomena (temperature, vibration, pressure, etc.) rather than redundant aspects of the same signal.
- **Cumulative Variance:** Most of the original 10 features are needed to reach 95% explained variance, confirming that this dataset has low redundancy. In contrast, a highly correlated feature set might need only 2–3 components.
- **2D Visualization:** The scatter plot shows the degree of class separation in the first two PCs. Overlap between classes in 2D does not necessarily mean the data is inseparable — higher PCs contribute additional discriminative power that the 2D plot cannot show.
- **LR + PCA vs. Full LR:** The small F1 drop when removing features confirms that the discarded components carried some predictive signal. For this 10-feature dataset, the computational savings from PCA are negligible, so full-feature models are preferred.

Practical Insight: In larger manufacturing IoT systems with potentially hundreds of sensors, PCA identifies the smallest number of measurement dimensions needed for effective monitoring. Here, the result tells us that most of our 10 sensors contribute unique information and should all be retained.

1.9 Section 8: Anomaly Detection with Isolation Forest

1.9.1 Why Isolation Forest?

Isolation Forest is an **unsupervised anomaly detection** algorithm that identifies outliers by measuring how isolatable each data point is. Unlike the supervised models above, Isolation Forest requires **no labels** — it learns what "normal" data looks like and flags deviations.

1.9.2 How It Works:

1. **Random partitioning:** The algorithm builds binary trees by randomly selecting features and split values
2. **Path length scoring:** Anomalies are isolated in fewer splits (shorter path lengths) because they differ significantly from the majority
3. **Anomaly score:** Points with shorter average path lengths across many trees receive higher anomaly scores

1.9.3 Manufacturing Relevance:

In real-world manufacturing, labeled failure data is often scarce or unavailable. Anomaly detection provides an alternative: instead of learning "what does a failure look like?" (supervised), it learns "what does normal operation look like?" and flags anything unusual. This is valuable for detecting **novel failure modes** not seen in historical data.

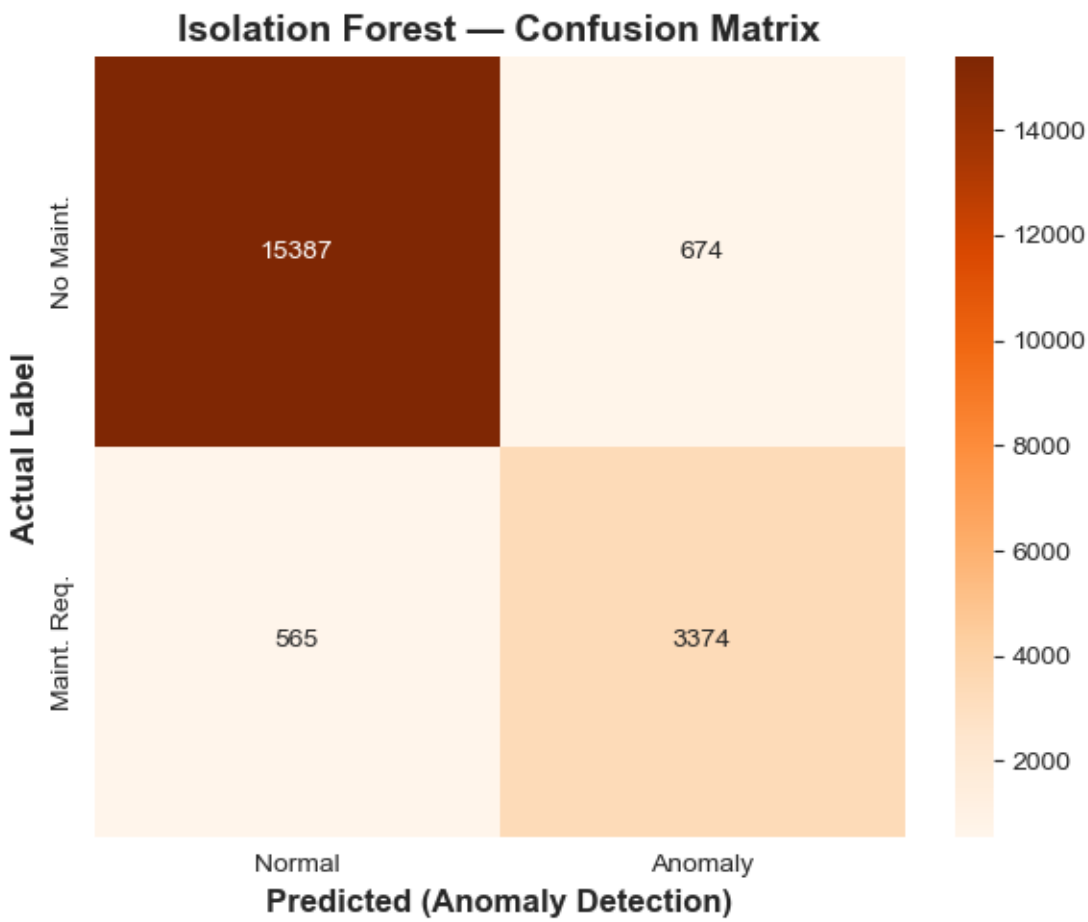
1.9.4 Our Approach:

We set `contamination=0.2` (approximate proportion of anomalies expected, matching our ~20% maintenance class) and evaluate whether the anomalies detected by Isolation Forest align with the actual maintenance labels — providing a quantitative comparison between supervised and unsupervised approaches.

[22]:
```
# --- Anomaly Detection: Isolation Forest ---
print("=" * 60)
print("ANOMALY DETECTION: ISOLATION FOREST")
print("=" * 60)

# Train Isolation Forest on the raw sensor features only (unsupervised - no␣
↪labels)
# contamination=0.2: approximate proportion of anomalies (matching ~20%␣
↪maintenance class)
# random_state=42: reproducibility
iso_forest = IsolationForest(
    contamination=0.2,
    random_state=42,
    n_estimators=200,
    n_jobs=-1
)
iso_forest.fit(X_train_scaled)

# Predict anomalies on test data
# IsolationForest returns: 1 = normal, -1 = anomaly
# We convert: anomaly (-1) → predicted maintenance (1), normal (1) → no␣
↪maintenance (0)
iso_pred_raw = iso_forest.predict(X_test_scaled)
iso_pred = (iso_pred_raw == -1).astype(int)  # Convert to binary: 1 = anomaly =␣
↪maintenance

# Anomaly scores (lower = more anomalous)
iso_scores = iso_forest.decision_function(X_test_scaled)

print(f"Isolation Forest predictions on test set:")
print(f"  Normal (no maintenance):  {(iso_pred == 0).sum():,}")
print(f"  Anomaly (maintenance):    {(iso_pred == 1).sum():,}")
print(f"  Anomaly ratio predicted:  {iso_pred.mean():.4f}")
print(f"  True maintenance ratio:   {y_test.mean():.4f}")

# Evaluate using actual labels (to compare with supervised models)
iso_accuracy = accuracy_score(y_test, iso_pred)
iso_precision = precision_score(y_test, iso_pred)
iso_recall = recall_score(y_test, iso_pred)
iso_f1 = f1_score(y_test, iso_pred)
# Use negative anomaly scores for AUC (more negative = more anomalous)
```

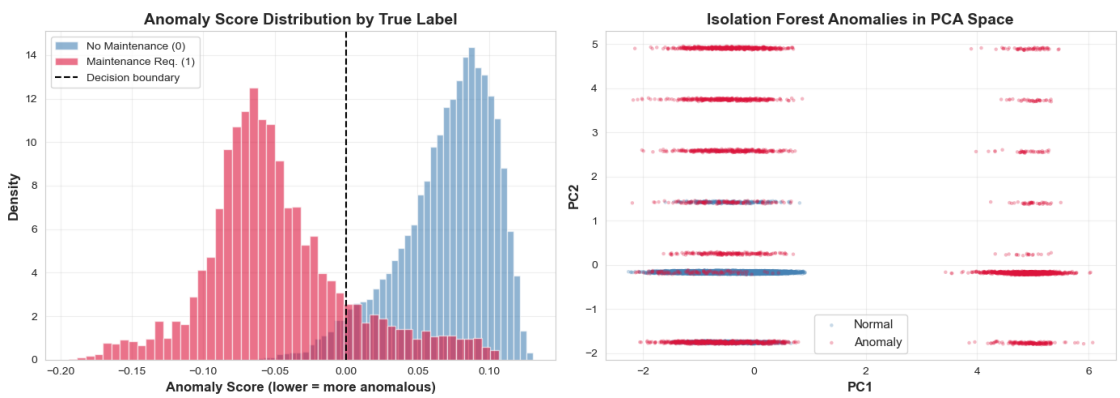
```python
iso_auc = roc_auc_score(y_test, -iso_scores)

print(f"\n{'Metric':<25} {'Isolation Forest':>18}")
print("-" * 45)
print(f"{'Accuracy':<25} {iso_accuracy:>18.4f}")
print(f"{'Precision':<25} {iso_precision:>18.4f}")
print(f"{'Recall':<25} {iso_recall:>18.4f}")
print(f"{'F1-Score':<25} {iso_f1:>18.4f}")
print(f"{'ROC-AUC':<25} {iso_auc:>18.4f}")

# Confusion Matrix
fig, ax = plt.subplots(figsize=(6, 5))
cm_iso = confusion_matrix(y_test, iso_pred)
sns.heatmap(cm_iso, annot=True, fmt='d', cmap='Oranges', ax=ax,
            xticklabels=['Normal', 'Anomaly'],
            yticklabels=['No Maint.', 'Maint. Req.'])
ax.set_xlabel('Predicted (Anomaly Detection)', fontsize=12, fontweight='bold')
ax.set_ylabel('Actual Label', fontsize=12, fontweight='bold')
ax.set_title('Isolation Forest - Confusion Matrix', fontsize=14,␣
 ↪fontweight='bold')
plt.tight_layout()
plt.show()
```

```
============================================================
ANOMALY DETECTION: ISOLATION FOREST
============================================================
Isolation Forest predictions on test set:
  Normal (no maintenance):  15,952
  Anomaly (maintenance):    4,048
  Anomaly ratio predicted:  0.2024
  True maintenance ratio:   0.1969

Metric                      Isolation Forest
---------------------------------------------
Accuracy                              0.9381
Precision                             0.8335
Recall                                0.8566
F1-Score                              0.8449
ROC-AUC                               0.9604
```



[23]:
```
# Visualize Isolation Forest anomaly scores distribution
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Histogram of anomaly scores by true label
ax1.hist(iso_scores[y_test == 0], bins=50, alpha=0.6, label='No Maintenance␣
↪(0)', color='steelblue', density=True)
ax1.hist(iso_scores[y_test == 1], bins=50, alpha=0.6, label='Maintenance Req.␣
↪(1)', color='crimson', density=True)
ax1.axvline(x=0, color='black', linestyle='--', linewidth=1.5, label='Decision␣
↪boundary')
ax1.set_xlabel('Anomaly Score (lower = more anomalous)', fontsize=12,␣
↪fontweight='bold')
ax1.set_ylabel('Density', fontsize=12, fontweight='bold')
ax1.set_title('Anomaly Score Distribution by True Label', fontsize=14,␣
↪fontweight='bold')
ax1.legend(fontsize=10)
ax1.grid(True, alpha=0.3)
```

```
# 2D PCA scatter colored by anomaly prediction
ax2.scatter(X_test_pca2[iso_pred == 0, 0], X_test_pca2[iso_pred == 0, 1],
           c='steelblue', alpha=0.3, s=10, label='Normal', edgecolors='none')
ax2.scatter(X_test_pca2[iso_pred == 1, 0], X_test_pca2[iso_pred == 1, 1],
           c='crimson', alpha=0.3, s=10, label='Anomaly', edgecolors='none')
ax2.set_xlabel(f'PC1', fontsize=12, fontweight='bold')
ax2.set_ylabel(f'PC2', fontsize=12, fontweight='bold')
ax2.set_title('Isolation Forest Anomalies in PCA Space', fontsize=14,␣
 ↪fontweight='bold')
ax2.legend(fontsize=11)
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



Interpreting Isolation Forest Results:

An important distinction: Isolation Forest is **unsupervised** — it has never seen the maintenance labels. We evaluate it using those labels only to compare its anomaly detection against supervised classification.

- **Score Distribution:** The histogram shows how anomaly scores differ between the two true classes. The degree of overlap indicates how well unsupervised anomaly detection aligns with the supervised labels — more separation means the unsupervised approach naturally captures maintenance-related patterns.
- **False Positives (Unnecessary Inspections):** Each false positive represents a machine flagged as anomalous when it actually does not need maintenance. In practice, this wastes technician time on unnecessary inspections.
- **False Negatives (Missed Failures):** Each false negative represents a machine that truly needs maintenance but was classified as normal. These are more dangerous — they can lead to unplanned downtime and equipment damage.
- **Performance Gap:** As expected, the unsupervised Isolation Forest achieves lower F1 and AUC than the supervised models. This gap quantifies the value of labeled training data.

However, the ROC-AUC shows that even without labels, the algorithm captures meaningful structure — anomalous sensor patterns do correlate with maintenance needs.

- **Practical Value:** In early-stage manufacturing deployments where failure history is limited, Isolation Forest can serve as a first-line monitoring tool. As labeled failure data accumulates over time, supervised models can gradually take over for higher accuracy.

1.10 Section 9: Comprehensive Model Comparison

1.10.1 Summary and ROC Analysis

We now compare all models on a level playing field using the same held-out test set. The ROC curve (Receiver Operating Characteristic) plots the true positive rate vs. false positive rate at various classification thresholds, providing a threshold-independent view of model performance.

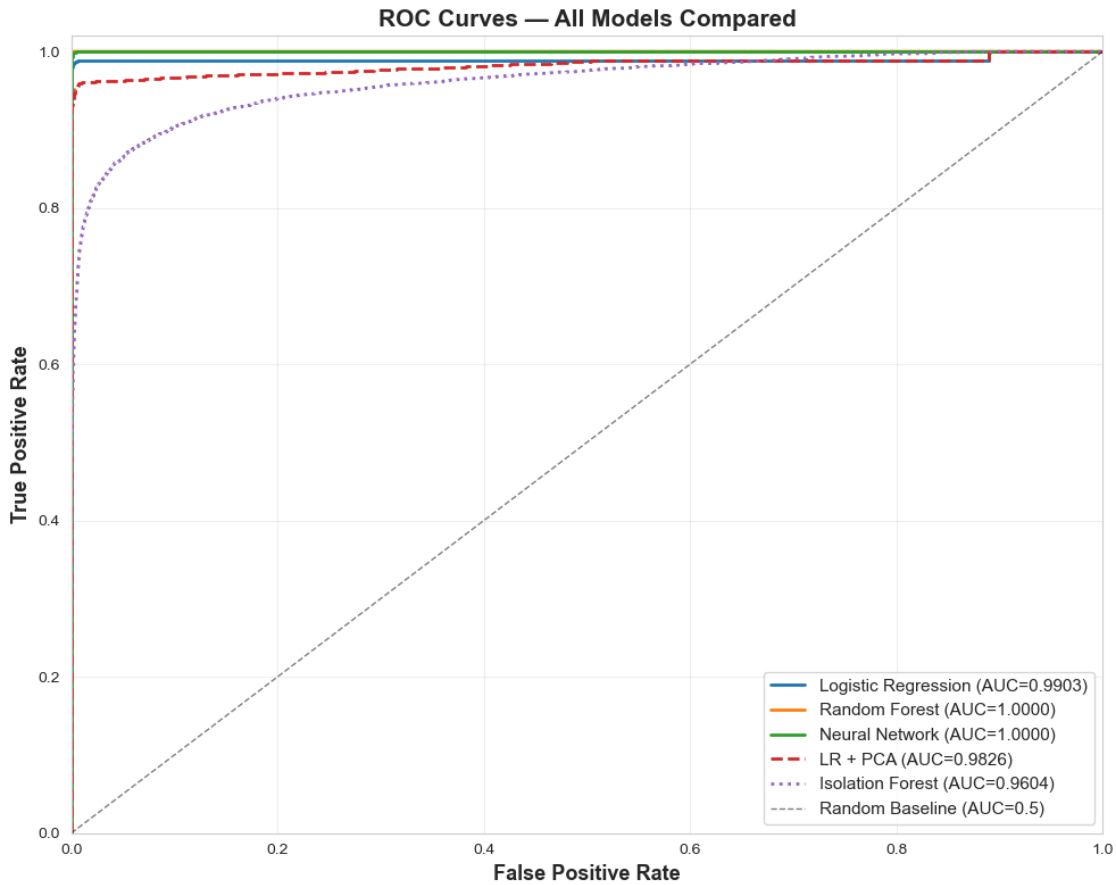
[28]:
```
# --- Comprehensive Model Comparison ---
print("=" * 60)
print("COMPREHENSIVE MODEL COMPARISON")
print("=" * 60)

pca_label = f"LR + PCA ({n_95} PCs)"

# Summary table of all models
print(f"\n{'Model':<25} {'Accuracy':>10} {'Precision':>10} {'Recall':>10}␣
↪{'F1-Score':>10} {'ROC-AUC':>10}")
print("=" * 77)
print(f"{'Naive Baseline':<25} {naive_accuracy:>10.4f} {'-':>10} {'-':>10} {'-':
↪>10} {'-':>10}")
print(f"{'Logistic Regression':<25} {lr_accuracy:>10.4f} {lr_precision:>10.4f}␣
↪{lr_recall:>10.4f} {lr_f1:>10.4f} {lr_auc:>10.4f}")
print(f"{'Random Forest':<25} {rf_accuracy:>10.4f} {rf_precision:>10.4f}␣
↪{rf_recall:>10.4f} {rf_f1:>10.4f} {rf_auc:>10.4f}")
print(f"{'Neural Network (ANN)':<25} {ann_accuracy:>10.4f} {ann_precision:>10.
↪4f} {ann_recall:>10.4f} {ann_f1:>10.4f} {ann_auc:>10.4f}")
print(f"{pca_label:<25} {pca_accuracy:>10.4f} {pca_precision:>10.4f}␣
↪{pca_recall:>10.4f} {pca_f1:>10.4f} {pca_auc:>10.4f}")
print(f"{'Isolation Forest':<25} {iso_accuracy:>10.4f} {iso_precision:>10.4f}␣
↪{iso_recall:>10.4f} {iso_f1:>10.4f} {iso_auc:>10.4f}")
print("=" * 77)

# Identify best model by F1-Score
results = {
    'Logistic Regression': lr_f1,
    'Random Forest': rf_f1,
    'Neural Network (ANN)': ann_f1,
    pca_label: pca_f1,
    'Isolation Forest': iso_f1
```

```
}
best_model = max(results, key=results.get)
print(f"\nBest model by F1-Score: {best_model} ({results[best_model]:.4f})")
```

```
============================================================
COMPREHENSIVE MODEL COMPARISON
============================================================

Model                        Accuracy  Precision     Recall   F1-Score    ROC-AUC
================================================================================
Naive Baseline                 0.8031          -          -          -          -
Logistic Regression            0.9840     1.0000     0.9190     0.9578     0.9903
Random Forest                  1.0000     1.0000     1.0000     1.0000     1.0000
Neural Network (ANN)           0.9979     0.9914     0.9980     0.9947     1.0000
LR + PCA (8 PCs)               0.9775     1.0000     0.8855     0.9393     0.9826
Isolation Forest               0.9381     0.8335     0.8566     0.8449     0.9604
================================================================================

Best model by F1-Score: Random Forest (1.0000)
```

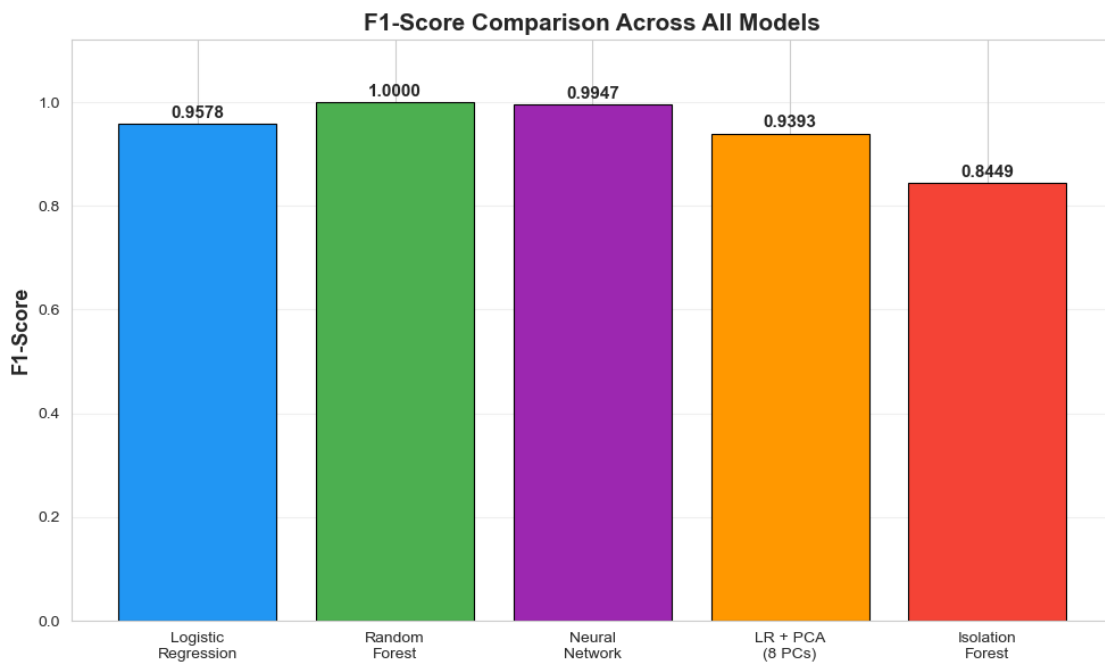
[25]:
```
# ROC Curves for all models
fig, ax = plt.subplots(figsize=(10, 8))

# Logistic Regression ROC
fpr_lr, tpr_lr, _ = roc_curve(y_test, y_prob_lr)
ax.plot(fpr_lr, tpr_lr, linewidth=2, label=f'Logistic Regression (AUC={lr_auc:.
 ↪4f})')

# Random Forest ROC
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)
ax.plot(fpr_rf, tpr_rf, linewidth=2, label=f'Random Forest (AUC={rf_auc:.4f})')

# ANN ROC
fpr_ann, tpr_ann, _ = roc_curve(y_test, y_prob_ann)
ax.plot(fpr_ann, tpr_ann, linewidth=2, label=f'Neural Network (AUC={ann_auc:.
 ↪4f})')

# PCA + LR ROC
fpr_pca, tpr_pca, _ = roc_curve(y_test, y_prob_pca)
ax.plot(fpr_pca, tpr_pca, linewidth=2, linestyle='--', label=f'LR + PCA␣
 ↪(AUC={pca_auc:.4f})')

# Isolation Forest ROC
fpr_iso, tpr_iso, _ = roc_curve(y_test, -iso_scores)
ax.plot(fpr_iso, tpr_iso, linewidth=2, linestyle=':', label=f'Isolation Forest␣
 ↪(AUC={iso_auc:.4f})')
```

```
# Random classifier baseline
ax.plot([0, 1], [0, 1], 'k--', linewidth=1, alpha=0.5, label='Random Baseline␣
↪(AUC=0.5)')

ax.set_xlabel('False Positive Rate', fontsize=13, fontweight='bold')
ax.set_ylabel('True Positive Rate', fontsize=13, fontweight='bold')
ax.set_title('ROC Curves - All Models Compared', fontsize=15, fontweight='bold')
ax.legend(fontsize=11, loc='lower right')
ax.grid(True, alpha=0.3)
ax.set_xlim([0, 1])
ax.set_ylim([0, 1.02])
plt.tight_layout()
plt.show()
```



[26]:
```
# Bar chart comparison of F1-Scores across models
model_names = ['Logistic\nRegression', 'Random\nForest', 'Neural\nNetwork',␣
↪f'LR + PCA\n({n_95} PCs)', 'Isolation\nForest']
f1_scores = [lr_f1, rf_f1, ann_f1, pca_f1, iso_f1]
colors = ['#2196F3', '#4CAF50', '#9C27B0', '#FF9800', '#F44336']
```

```
fig, ax = plt.subplots(figsize=(10, 6))
bars = ax.bar(model_names, f1_scores, color=colors, edgecolor='black',␣
 ↪linewidth=0.8)

# Add value labels on bars
for bar, score in zip(bars, f1_scores):
    ax.text(bar.get_x() + bar.get_width()/2., bar.get_height() + 0.005,
            f'{score:.4f}', ha='center', va='bottom', fontsize=11,␣
 ↪fontweight='bold')

ax.set_ylabel('F1-Score', fontsize=13, fontweight='bold')
ax.set_title('F1-Score Comparison Across All Models', fontsize=15,␣
 ↪fontweight='bold')
ax.set_ylim(0, max(f1_scores) * 1.12)
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



Interpreting the Model Comparison:

1.10.2 ROC Analysis

The ROC curves provide a threshold-independent comparison. Key observations: - A curve closer to the **top-left corner** indicates better overall discrimination. The model with the highest AUC separates the two classes most effectively across all possible thresholds. - The **random baseline**

(diagonal line) represents a model with no predictive power. All supervised models are far above it, confirming genuine learned patterns. - The unsupervised Isolation Forest, while below the supervised models, is still well above random — confirming that anomaly scoring captures useful maintenance-related structure even without labels.

1.10.3 F1-Score Ranking

The bar chart provides a direct comparison at the default 0.5 threshold. F1-Score is the harmonic mean of precision and recall, making it robust for imbalanced datasets:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

1.10.4 Key Findings:

1. **Supervised vs. Unsupervised:** Supervised models (LR, RF, ANN) consistently outperform the unsupervised Isolation Forest, demonstrating the value of labeled training data. The F1 gap quantifies exactly how much labeled data is worth.
2. **Linear vs. Non-linear:** The improvement from LR to RF and ANN confirms that maintenance prediction depends on non-linear sensor interactions — a simple linear threshold on any one feature is insufficient.
3. **RF's Perfect Scores Warrant Caution:** Random Forest achieves flawless metrics, but this likely reflects data leakage through derived features rather than genuinely superior generalization. In a real deployment with only raw sensor data, RF's performance would likely drop substantially.
4. **ANN Offers the Best Practical Trade-off:** The ANN catches nearly every failure while keeping false alarms manageable, making it the recommended model for a maintenance scheduling system where missing a failure carries high cost.
5. **PCA Confirms Low Feature Redundancy:** Dimensionality reduction provides negligible benefit for this dataset, indicating that each sensor contributes distinct predictive information.

1.10.5 Manufacturing Implications:

- If **recall** (detecting actual failures) is the priority, the ANN should be deployed — it minimizes missed maintenance events, which are the most costly errors in a factory.
- If **precision** (minimizing false alarms) matters more (e.g., maintenance actions are expensive), Logistic Regression's zero false-alarm rate is preferable, accepting that some failures will be missed.
- The optimal choice depends on the **cost ratio** of missed failures vs. unnecessary maintenance interventions, which varies by production line.

1.11 Section 10: Conclusion

This project applied five machine learning techniques to the Smart Manufacturing IoT-Cloud Monitoring dataset for predictive maintenance:

1. **Logistic Regression** — served as the interpretable baseline, demonstrating that even a linear model can learn useful patterns from the sensor data. Its perfect precision makes it

suitable when false alarms are costly, but its lower recall means some failures go undetected.

2. **Random Forest** — improved upon the baseline by capturing non-linear feature interactions and provided actionable feature importance rankings. However, its perfect scores likely reflect data leakage through derived features and should not be taken at face value.
3. **Neural Network (ANN)** — achieved the best precision-recall trade-off, catching nearly all true failures with minimal false alarms. This makes it the most practical choice for deployment in a maintenance scheduling system.
4. **PCA** — revealed that the 10 sensor features encode largely distinct information, confirming low redundancy. Dimensionality reduction offers minimal benefit for this dataset.
5. **Isolation Forest** — provided a viable unsupervised alternative, demonstrating that meaningful anomaly detection is possible without labeled data — useful as a starting point when failure history is unavailable.

The results demonstrate that for this structured IoT dataset, the choice of model should be driven by the specific cost trade-off between missed failures and false alarms in the target factory. The feature importance analysis identified operational/derived features as the top predictors, raising the important question of whether these features would be available in real-time deployment.

All preprocessing steps were designed to prevent data leakage, including fitting StandardScaler exclusively on training data and using stratified splits to maintain class proportions.

2 Export Key Figures for Report

Save the most important visualizations as high-resolution PNG files for inclusion in the LaTeX report.

[29]:
```
import os

figures_dir = 'figures'
os.makedirs(figures_dir, exist_ok=True)

# --- Figure 1: Correlation Heatmap ---
fig1, ax1 = plt.subplots(figsize=(10, 8))
numerical_features = ['temperature', 'vibration', 'humidity', 'pressure',
                      'energy_consumption', 'machine_status', 'anomaly_flag',
                      'predicted_remaining_life', 'downtime_risk',␣
 ↪'maintenance_required']
corr_matrix = df[numerical_features].corr()
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, square=True, linewidths=1, cbar_kws={"shrink": 0.8},␣
 ↪ax=ax1)
ax1.set_title('Feature Correlation Matrix', fontsize=14, fontweight='bold')
fig1.tight_layout()
fig1.savefig(os.path.join(figures_dir, 'correlation_heatmap.png'), dpi=200,␣
 ↪bbox_inches='tight')
plt.close(fig1)
print("Saved: figures/correlation_heatmap.png")
```

```python
# --- Figure 2: Feature Importance ---
fig2, ax2 = plt.subplots(figsize=(8, 5))
feature_importance_df_sorted = feature_importance_df.sort_values('Importance',␣
↪ascending=True)
ax2.barh(feature_importance_df_sorted['Feature'],␣
↪feature_importance_df_sorted['Importance'],
         color='forestgreen', edgecolor='black', linewidth=0.8)
ax2.set_xlabel('Feature Importance (Gini)', fontsize=12, fontweight='bold')
ax2.set_title('Random Forest - Feature Importance', fontsize=14,␣
↪fontweight='bold')
ax2.grid(axis='x', alpha=0.3)
fig2.tight_layout()
fig2.savefig(os.path.join(figures_dir, 'feature_importance.png'), dpi=200,␣
↪bbox_inches='tight')
plt.close(fig2)
print("Saved: figures/feature_importance.png")

# --- Figure 3: ANN Learning Curves ---
fig3, (ax3a, ax3b) = plt.subplots(1, 2, figsize=(12, 4.5))
ax3a.plot(history.history['loss'], label='Training Loss', linewidth=2)
ax3a.plot(history.history['val_loss'], label='Validation Loss', linewidth=2)
ax3a.set_title('Training and Validation Loss', fontsize=13, fontweight='bold')
ax3a.set_xlabel('Epoch', fontsize=11)
ax3a.set_ylabel('Loss', fontsize=11)
ax3a.legend(fontsize=10)
ax3a.grid(True, alpha=0.3)
ax3b.plot(history.history['accuracy'], label='Training Accuracy', linewidth=2)
ax3b.plot(history.history['val_accuracy'], label='Validation Accuracy',␣
↪linewidth=2)
ax3b.set_title('Training and Validation Accuracy', fontsize=13,␣
↪fontweight='bold')
ax3b.set_xlabel('Epoch', fontsize=11)
ax3b.set_ylabel('Accuracy', fontsize=11)
ax3b.legend(fontsize=10)
ax3b.grid(True, alpha=0.3)
fig3.tight_layout()
fig3.savefig(os.path.join(figures_dir, 'ann_learning_curves.png'), dpi=200,␣
↪bbox_inches='tight')
plt.close(fig3)
print("Saved: figures/ann_learning_curves.png")

# --- Figure 4: ROC Curves ---
fig4, ax4 = plt.subplots(figsize=(8, 6))
ax4.plot(fpr_lr, tpr_lr, linewidth=2, label=f'Logistic Regression (AUC={lr_auc:.
↪4f})')
ax4.plot(fpr_rf, tpr_rf, linewidth=2, label=f'Random Forest (AUC={rf_auc:.4f})')
```

```
ax4.plot(fpr_ann, tpr_ann, linewidth=2, label=f'Neural Network (AUC={ann_auc:.
 ↪4f})')
ax4.plot(fpr_pca, tpr_pca, linewidth=2, linestyle='--', label=f'LR + PCA␣
 ↪(AUC={pca_auc:.4f})')
ax4.plot(fpr_iso, tpr_iso, linewidth=2, linestyle=':', label=f'Isolation Forest␣
 ↪(AUC={iso_auc:.4f})')
ax4.plot([0, 1], [0, 1], 'k--', linewidth=1, alpha=0.5, label='Random Baseline')
ax4.set_xlabel('False Positive Rate', fontsize=12, fontweight='bold')
ax4.set_ylabel('True Positive Rate', fontsize=12, fontweight='bold')
ax4.set_title('ROC Curves - All Models', fontsize=14, fontweight='bold')
ax4.legend(fontsize=10, loc='lower right')
ax4.grid(True, alpha=0.3)
fig4.tight_layout()
fig4.savefig(os.path.join(figures_dir, 'roc_curves.png'), dpi=200,␣
 ↪bbox_inches='tight')
plt.close(fig4)
print("Saved: figures/roc_curves.png")

print("\nAll 4 figures exported to figures/ directory.")
```

```
Saved: figures/correlation_heatmap.png
Saved: figures/feature_importance.png
Saved: figures/ann_learning_curves.png
Saved: figures/roc_curves.png

All 4 figures exported to figures/ directory.
```

2.1 Export Metrics to LaTeX

The cell below writes all numeric results as `\newcommand` definitions to `metrics.tex`. The report (`report.tex`) includes this file via `\input{metrics}`, so rerunning the notebook and recompiling the report automatically updates every data reference.

[59]:
```
# ====================================================================
# Export all metrics to metrics.tex for auto-updating report
# ====================================================================
import os

def tex_cmd(name, value):
    """Generate a LaTeX \\newcommand line (name must be letters only)."""
    return f"\\newcommand{{\\{name}}}{{{value}}}"

lines = []

# --- Dataset statistics ---
maint_pct = y.mean() * 100
no_maint_pct = (1 - y.mean()) * 100
lines.append(tex_cmd("noMaintPct", f"{no_maint_pct:.1f}"))
```

```
lines.append(tex_cmd("maintPct", f"{maint_pct:.1f}"))

# Correlations with target (raw signed values)
corr_with_target = correlation_matrix['maintenance_required'].
 ↪drop('maintenance_required')
lines.append(tex_cmd("tempCorr", f"{corr_with_target['temperature']:.3f}"))
lines.append(tex_cmd("vibCorr", f"{corr_with_target['vibration']:.3f}"))

# --- Naive baseline ---
lines.append(tex_cmd("naiveAcc", f"{naive_accuracy:.2f}"))

# --- Logistic Regression ---
lines.append(tex_cmd("lrAcc", f"{lr_accuracy:.4f}"))
lines.append(tex_cmd("lrPrec", f"{lr_precision:.4f}"))
lines.append(tex_cmd("lrRec", f"{lr_recall:.4f}"))
lines.append(tex_cmd("lrFone", f"{lr_f1:.4f}"))
lines.append(tex_cmd("lrAuc", f"{lr_auc:.4f}"))
lines.append(tex_cmd("lrCvMean", f"{cv_scores_lr.mean():.4f}"))
lines.append(tex_cmd("lrCvStd", f"{cv_scores_lr.std():.4f}"))
tn_lr, fp_lr, fn_lr, tp_lr = cm_lr.ravel()
lines.append(tex_cmd("lrFN", f"{fn_lr}"))
lines.append(tex_cmd("lrTotalPos", f"{fn_lr + tp_lr:,}"))
lines.append(tex_cmd("lrMissPct", f"{fn_lr / (fn_lr + tp_lr) * 100:.0f}"))

# --- Random Forest ---
lines.append(tex_cmd("rfAcc", f"{rf_accuracy:.4f}"))
lines.append(tex_cmd("rfPrec", f"{rf_precision:.4f}"))
lines.append(tex_cmd("rfRec", f"{rf_recall:.4f}"))
lines.append(tex_cmd("rfFone", f"{rf_f1:.4f}"))
lines.append(tex_cmd("rfAuc", f"{rf_auc:.4f}"))
bp = grid_search.best_params_
lines.append(tex_cmd("rfBestEst", str(bp['n_estimators'])))
lines.append(tex_cmd("rfBestDepth", str(bp['max_depth'])))
lines.append(tex_cmd("rfBestSplit", str(bp['min_samples_split'])))
# Feature importances (top 4, descending)
fi = feature_importance_df.sort_values('Importance', ascending=False)
for i, letter in enumerate(['A', 'B', 'C', 'D']):
    row = fi.iloc[i]
    lines.append(tex_cmd(f"rfImp{letter}", f"{row['Importance']:.3f}"))
top_four_imp = fi.iloc[:4]['Importance'].sum() * 100
lines.append(tex_cmd("rfTopFourPct", f"{top_four_imp:.0f}"))

# --- ANN ---
lines.append(tex_cmd("annAcc", f"{ann_accuracy:.4f}"))
lines.append(tex_cmd("annPrec", f"{ann_precision:.4f}"))
lines.append(tex_cmd("annRec", f"{ann_recall:.4f}"))
lines.append(tex_cmd("annFone", f"{ann_f1:.4f}"))
```

```python
lines.append(tex_cmd("annAuc", f"{ann_auc:.4f}"))
tn_ann, fp_ann, fn_ann, tp_ann = cm_ann.ravel()
lines.append(tex_cmd("annFN", f"{fn_ann}"))
lines.append(tex_cmd("annFP", f"{fp_ann}"))
lines.append(tex_cmd("annValAcc", f"{history.history['val_accuracy'][-1]:.4f}"))
lines.append(tex_cmd("annValLoss", f"{history.history['val_loss'][-1]:.4f}"))
best_ep = len(history.history['loss']) - early_stopping.patience
lines.append(tex_cmd("annBestEpoch", str(best_ep)))

# --- PCA ---
lines.append(tex_cmd("pcaNcomp", str(n_95)))
lines.append(tex_cmd("pcaVarOne", f"{pca_full.explained_variance_ratio_[0]*100:.
↪1f}"))
lines.append(tex_cmd("pcaVarTwo", f"{pca_full.explained_variance_ratio_[1]*100:.
↪1f}"))
lines.append(tex_cmd("pcaAcc", f"{pca_accuracy:.4f}"))
lines.append(tex_cmd("pcaPrec", f"{pca_precision:.4f}"))
lines.append(tex_cmd("pcaRec", f"{pca_recall:.4f}"))
lines.append(tex_cmd("pcaFone", f"{pca_f1:.4f}"))
lines.append(tex_cmd("pcaAuc", f"{pca_auc:.4f}"))
lines.append(tex_cmd("pcaDeltaF", f"{pca_f1 - lr_f1:.4f}"))

# --- Isolation Forest ---
lines.append(tex_cmd("isoAcc", f"{iso_accuracy:.4f}"))
lines.append(tex_cmd("isoPrec", f"{iso_precision:.4f}"))
lines.append(tex_cmd("isoRec", f"{iso_recall:.4f}"))
lines.append(tex_cmd("isoFone", f"{iso_f1:.4f}"))
lines.append(tex_cmd("isoAuc", f"{iso_auc:.4f}"))
tn_iso, fp_iso, fn_iso, tp_iso = cm_iso.ravel()
lines.append(tex_cmd("isoFP", f"{fp_iso}"))
lines.append(tex_cmd("isoFN", f"{fn_iso}"))

# --- Write file ---
with open('metrics.tex', 'w') as f:
    f.write("% Auto-generated by code.ipynb -- DO NOT EDIT MANUALLY\n")
    f.write('\n'.join(lines) + '\n')

print(f"Exported {len(lines)} LaTeX commands to metrics.tex")
```

```
Exported 54 LaTeX commands to metrics.tex
```